

ALGERIA

NATIONAL CLIMATE PLAN

English Translation

Source document dated September 2019

الجمهورية الجزائرية الديمقراطية الشعبية
République Algérienne Démocratique et Populaire



وزارة البيئة و الطاقات المتجددة
Ministère de l'Environnement et des Energies Renouvelables



People's Democratic Republic of Algeria
Ministry of Environment and Renewable Energy

Non-certified translation

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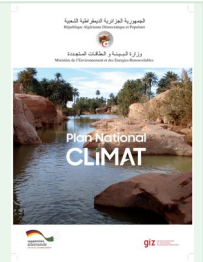
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PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA

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PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA

MINISTRY OF ENVIRONMENT AND RENEWABLE ENERGY

NATIONAL CLIMATE PLAN

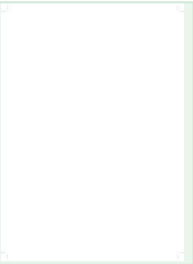
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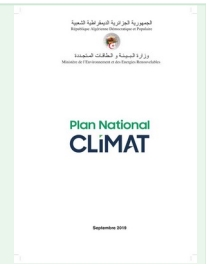
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SOURCE PDF PAGE 03**PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA**

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**PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA****MINISTRY OF ENVIRONMENT AND RENEWABLE ENERGY****NATIONAL CLIMATE PLAN**

September 2019

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FOREWORD

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FOREWORD

The preparation of Algeria's National Climate Plan forms part of a set of actions initiated by Algeria under the country's national development strategy, as set out in the Government's action program. It complements other plans, including the National Action Plan for the Environment and Sustainable Development (PNAEDD), the National Biodiversity Strategy and Action Plan (SPANB), and the National Integrated Waste Management Strategy (SNGID) through 2035.

Completion of this plan was made possible by the commitment and active contribution of the various sectors through working groups and interim meetings, under the oversight and with the support of the National Climate Committee (CNC), established in July 2015 by order of the Prime Minister.

This plan brings together all actions planned or envisaged by the sectors over different time horizons, covering both adaptation to climate change and the mitigation of greenhouse-gas emissions.

The participatory and inclusive approach adopted in carrying out the work not only strengthened sectoral capacities, but also enabled consultation within each sector and promoted intersectoral cooperation on climate-change issues.

The NCP includes some of the actions that Algeria must implement to honor the commitments expressed in its Nationally Determined Contribution (NDC) to reduce greenhouse-gas emissions. Following the Paris Agreement on climate change, Algeria committed in its NDC to reduce its greenhouse-gas emissions by 7% using its own resources, and by as much as 22% with international financial and technological support.

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FOREWORD - CONTINUED

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**FOREWORD - CONTINUED**

Mitigation actions cover the energy, housing, transport, industry, and forestry sectors, including actions undertaken by local authorities and the private sector. Adaptation actions concern water resources, agriculture, forests, health, and the coastal zone, and also include actions at the local level.

The Nationally Determined Contribution (NDC) is an opportunity to develop energy efficiency, rationalize energy consumption, and expand renewable energy, thereby reducing greenhouse-gas emissions. At the same time, this will protect national energy resources; improve the performance of industrial processes and energy-management methods; modernize facilities; and initiate a change in everyone's behavior toward energy derived from fossil fuels. Ultimately, implementation of the NCP contributes to Algeria's energy transition toward sustainable energy.

Algeria's vulnerability to climate change has been demonstrated in all studies conducted in strategic areas such as food security and the population's water supply, as well as the risk of flooding in urban areas and the weakening of basic infrastructure. Climate-change adaptation actions constitute the other component of the NCP. Algeria will advance these actions using its own resources while seeking international support through climate finance.

This NCP is Algeria's first national showcase for climate action. For Algerians, it is intended to show sectors the way forward and encourage commitment to combating climate change. It is also a window to the international community, demonstrating Algeria's implementation of its commitments, whether under the United Nations Framework Convention on Climate Change (UNFCCC) or the Paris Agreement.

I wish to thank all members of the National Climate Committee (CNC), the sectoral working groups that were formed, and everyone who contributed directly or indirectly to this work, for all the effort devoted over more than a year of intensive work to preparing the National Climate Plan (NCP). I personally followed the preparation process from its launch through its validation by the National Climate Committee (CNC).

This NCP is only the first cornerstone of the major undertaking to combat climate change in Algeria. In the future, it should be reviewed and updated periodically to take account of the country's future socioeconomic context, new scientific knowledge, and developments in international negotiations.

I hope to see its implementation begin very soon across all sectors, in full synergy and with the necessary intra- and intersectoral coordination to ensure its success.

Ms. Fatma Zohra Zerouati

Minister of Environment and Renewable Energy

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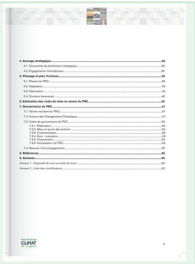


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ABBREVIATIONS

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ABBREVIATIONS

AADC	- Association for Community Development Action
AFEV	- Association of Women in the Green Economy
AGIRE	- National Agency for Integrated Water Resources Management
ANCC	- National Climate Change Agency
AND	- National Waste Agency
ANN	- National Agency for Nature Conservation
ANRH	- National Water Resources Agency
AP	- Protected Area
APRUE	- National Agency for the Promotion and Rationalization of Energy Use
ARV	- Climate-Change Risk and Vulnerability Analysis
ASAL	- Algerian Space Agency
BASIC	- Group of countries comprising Brazil, South Africa, India, and China
BAU	- "Business as Usual" scenario, or baseline scenario
BHC	- Municipal Hygiene Office
BM	- World Bank
BRICS	- Group of countries comprising Brazil, Russia, India, China, and South Africa
BUR	- Biennial Update Report
CC	- Climate Change
CCNUCC	- United Nations Framework Convention on Climate Change
CDARS	- Commission for the Development of Agriculture in Saharan Regions
CDER	- Renewable Energy Development Center
CDN	- Nationally Determined Contribution
CPDN	- Intended Nationally Determined Contribution
CET	- Engineered Landfill Center
CNC	- National Climate Committee
CNERIB	- National Center for Integrated Building Studies and Research

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ABBREVIATIONS - CONTINUED

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ABBREVIATIONS - CONTINUED

CNES - National Economic and Social Council

CNL - National Coastal Commission

CNRDPA - National Center for Fisheries and Aquaculture Research

CNT - National Toxicology Center

CNTPP - National Center for Cleaner Production Technologies

CNUCED - United Nations Conference on Trade, Environment, and Development

COMENA - Atomic Energy Commission

CoP - Conference of the Parties to the Framework Convention on Climate Change

CoP-MoP - Conference of the Parties serving as the Meeting of the Parties to the Kyoto Protocol

CRSTRA - Scientific and Technical Research Center for Arid Regions

CZH - Wetland Complex

DA - Algerian dinars

DCC - Climate Change Directorate

DGEDD - General Directorate for Environment and Sustainable Development

DGF - General Directorate of Forests

DGPA - General Directorate of Fisheries and Aquaculture

DGPS - General Delegation for Prevention and Security

DGUA - General Directorate of Urban Planning and Architecture

DNRM - National Delegation for Major Risks

DGRSDT - General Directorate for Scientific Research and Technological Development

DTP - Public Works Directorate

EGSA - Algiers Airport Services Management Establishment

ENPI - National Real Estate Development Enterprise

ENSSMAL - National Higher School of Marine Sciences and Coastal Planning

ETP - Potential evapotranspiration

ETPH - Hydraulic Public Works Enterprise

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ABBREVIATIONS - CONTINUED

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**ABBREVIATIONS - CONTINUED**

EVV	- Bougezoul City Establishment
EVNB	- Bouinan New City Establishment
FADHE	- Algerian Forum for Human Rights and the Environment
FEM	- Global Environment Facility
FNRSDT	- National Fund for Scientific Research and Technological Development
FVC	- Green Climate Fund
GES	- Greenhouse Gas
GICA	- Industrial Group of Algerian Cements
GIEC	- Intergovernmental Panel on Climate Change
HCDS	- High Commission for Steppe Development
IACC	- Integration of Climate-Change Adaptation
IHFR	- Hydrometeorological Training and Research Institute
INSP	- National Institute of Public Health
INRAA	- National Institute of Agronomic Research of Algeria
INRF	- National Forestry Research Institute
INSID	- National Institute of Soils, Irrigation, and Drainage
IPA	- Pasteur Institute of Algeria
ITAFV	- Technical Institute for Fruit Arboriculture and Viticulture
ITCMI	- Technical Institute for Vegetable and Industrial Crops
ITDAS	- Technical Institute for the Development of Saharan Agronomy
ITGC	- Technical Institute for Field Crops
LEM	- Maritime Studies Laboratory
MADRP	- Ministry of Agriculture, Rural Development, and Fisheries
MAE	- Ministry of Foreign Affairs
MC	- Ministry of Commerce
MDN	- Ministry of National Defense
MATE	- Ministry of Spatial Planning and the Environment

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ABBREVIATIONS - CONTINUED

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ABBREVIATIONS - CONTINUED

MEER	- Ministry of Environment and Renewable Energy
MEFP	- Ministry of Education and Vocational Training
ME	- Ministry of Energy
MEN	- Ministry of National Education
MESRS	- Ministry of Higher Education and Scientific Research
MF	- Ministry of Finance
MHUV	- Ministry of Housing, Urban Planning, and the City
MICLAT	- Ministry of the Interior, Local Authorities, and Spatial Planning
MIM	- Ministry of Industry and Mines
MSPRH	- Ministry of Health, Population, and Hospital Reform
MRE	- Ministry of Water Resources
MRV	- Measuring, Reporting, and Verification
MTH	- Waterborne diseases
MTPT	- Ministry of Public Works and Transport
MTA	- Ministry of Tourism and Handicrafts
MTV	- Vector-borne diseases
NAFTAL	- National Company for the Marketing and Distribution of Petroleum Products
NAFTEC	- National Petroleum Refining Company
NU	- United Nations
OAIC	- Algerian Interprofessional Cereals Office
ODD	- United Nations Sustainable Development Goals
ONEDD	- National Observatory for Environment and Sustainable Development
ONG	- Non-Governmental Organization
ONID	- National Irrigation and Drainage Office
ONM	- National Meteorological Office
ONS	- National Statistics Office
ONU	- United Nations Organization

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ABBREVIATIONS - CONTINUED

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ABBREVIATIONS - CONTINUED

OPGI	- Real Estate Promotion and Management Office
ORS	- Regional Health Observatory
PDAU	- Master Plan for Development and Urban Planning
PIB	- Gross Domestic Product
PK	- Kyoto Protocol
PNA	- National Adaptation Plan
PNAEDD	- National Action Plan for the Environment and Sustainable Development
PNC	- National Climate Plan
PNR	- National Research Programs
PNUD	- United Nations Development Programme
PNUE	- United Nations Environment Programme
POS	- Land-Use Plan
PPRI	- Flood-Risk Prevention Plan
PROGDEM	- Municipal Solid-Waste Management Program
MENA	- Middle East and North Africa Region
RCP	- Representative Concentration Pathway
RGA	- General Agricultural Census
RNE	- National Report on the State of the Environment
SAU	- Utilized agricultural area
SCZH	- Wetland sub-complex
SDAL	- National Coastal Development Plan
SIG	- Geographic Information System
SHFN	- Hydrographic Service of the Naval Forces
SNGID	- National Integrated Waste Management Strategy
SNAT	- National Spatial Planning Scheme
SNEDD	- National Strategy for Environment and Sustainable Development

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ABBREVIATIONS - CONTINUED

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ABBREVIATIONS - CONTINUED

SONATRACH - National Company for Hydrocarbon Research, Production, Transport, Processing, and Marketing

SONELGAZ - National Electricity and Gas Company

SPANB - National Biodiversity Strategy and Action Plan

SPI - Standardized Precipitation Index

SSAP - Early-Warning Monitoring System

STEP - Wastewater Treatment Plants

TCN - Third National Communication on Climate Change

TIC - Information and Communication Technologies

UICN - International Union for Conservation of Nature

UNISDR - United Nations Office for Disaster Risk Reduction

USD - United States dollars

VNSA - Sidi Abdellah New City

SOURCE PDF PAGE 16

EXECUTIVE SUMMARY

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**EXECUTIVE SUMMARY**

Like other countries in its region, Algeria is particularly affected by desertification and land degradation. Areas of the country receiving more than 400 mm of rainfall per year are confined to a strip extending no more than 150 km inland from the coast. Climate variability and extreme weather events pose serious challenges for the population, potentially increasing food insecurity and hindering the country's socioeconomic development.

The National Climate Plan (NCP) is the centerpiece of Algeria's strategic approach to countering the effects of climate change. It presents a set of actions that will contribute to achieving objectives for adapting to climate-change impacts and reducing greenhouse-gas (GHG) emissions.

The NCP is not Algeria's only instrument. Other strategies, policies, and guidelines will reinforce the efforts undertaken in this area over the coming years.

The NCP was designed to be consistent with the new economic growth model; the Government Action Plan; the National Spatial Planning Scheme (SNAT, 2017); environment-related strategies (the National Strategy for Environment and Sustainable Development and the National Biodiversity Strategy); and Algeria's international obligations, which include achieving the objective of Article 2 of the Convention on Climate Change. The NCP's objectives concern the identification of:

An adaptation action plan to address the effects of climate change, in order to: (i) adapt to warming that is twice the global average (a trend over nearly a century of 1.5°C compared with 0.7°C); and (ii) adapt to a decline in rainfall ranging from 20% in the west to 10% in the east.

A greenhouse-gas-emissions mitigation action plan that respects international commitments, in order to: (i) achieve the voluntary scenario of a 7% reduction by 2030; and (ii) achieve the conditional scenario of a 22% reduction by 2030.

The evolving nature of the NCP should be emphasized. Additional initiatives will be introduced during its implementation, depending on Algeria's socioeconomic progress and available budgets. In this regard, Phase 2 of the NCP, covering 2025-2030, will take such initiatives into account in order to strengthen their effects and align the phase with the Nationally Determined Contribution (NDC).

Implementation of the NCP requires the establishment of dynamic partnerships between the Government and key civil-society stakeholders. Achieving the reduction target is a major challenge to which Algerian society as a whole is called upon to contribute.

In this regard, the Government is establishing conditions that will enable the Algerian economy to benefit as fully as possible from the positive effects of measures to combat the impacts of climate change.

SOURCE PDF PAGE 17

1. INTRODUCTION

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1. INTRODUCTION

As the impacts of climate disruption multiply and pose a genuine threat to global prosperity, Algeria has adopted a voluntary approach to combating climate change and adapting to the country's new climatic conditions. The Government intends to accelerate implementation of the Paris Agreement, ratified by presidential decree¹ in 2016, thereby confirming Algeria's Nationally Determined Contribution (NDC).²

The NDC states that Algeria undertakes to reduce its GHG emissions by setting a national target of 7% below the reference, or Business as Usual (BAU), scenario by 2030 using its own resources, and 22% if the country receives international financing.

Accordingly, a process was launched to prepare the National Climate Plan (NCP), with the aim of translating the NDC objectives into concrete actions consistent with those of highly important strategic instruments.³ This process provides an opportunity to bring all Algerian stakeholders⁴ together around the responses needed to address the challenges raised by climate change.

The National Climate Plan (NCP) is the centerpiece of Algeria's strategic approach to countering the effects of climate change. It presents a set of actions that will contribute to achieving the objectives of reducing greenhouse-gas (GHG) emissions and adapting to climate-change impacts. The NCP is not Algeria's only instrument; other strategies, policies, and guidelines will reinforce the efforts undertaken in this area over the coming years.

This NCP report is structured as follows:

This introduction (Section 1).

Section 2 specifies the NCP's objectives, preparation methodology, principles, and implementation conditions.

Section 3 describes the Algerian climate-change context, presents the main impacts of climate change on the economy and society, describes the international context, and sets out Algeria's mitigation and adaptation objectives.

Section 4 presents the NCP's strategic anchoring.

Section 5 sets out the planned actions.

Section 6 presents an estimate of the investments required to implement the NCP.

Section 7 proposes the NCP governance framework.

Sections 8 and 9 contain the bibliographic references and the report's annexes.

The NCP consists of this report, fact sheets describing each planned action, the report annexes, and a policy brief for decision-makers.

¹ Ratification of the Climate Agreement by Presidential Decree No. 16-262 of 11 Muharram 1438, corresponding to 13 October 2016, concerning the agreement adopted in Paris on 12 December 2015, confirms Algeria's contribution.

² Following adoption of the Paris Agreement, all Parties' submitted Intended Nationally Determined Contributions (INDCs/CPDNs), unless updated after the initial submission, automatically became the countries' first Nationally Determined Contributions (NDCs/CDNs).

³ These are the Government Action Plan, the National Spatial Planning Scheme to 2030 (SNAT), the National Strategy for Environment and Sustainable Development (SNEDD), and the Sustainable Development Goals (SDGs), drawing on the SNEDD and the SNGID 2035.

⁴ Involvement of civil society and the private sector in both adaptation and mitigation efforts.

SOURCE PDF PAGE 18

2. OBJECTIVES AND METHODOLOGY

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2. OBJECTIVES AND METHODOLOGY

2.1. Objectives of the NCP

The objective of the NCP is to select and organize a set of proposed actions to enable the national economy to adapt to climate change and Algeria to achieve its GHG-mitigation objectives.

More specifically, the NCP aims to:

- Identify, through the participation of institutional sectors and civil society, the mitigation and adaptation actions to be implemented.
- Determine which actions should be implemented as a priority in the short term.
- Propose a governance framework by identifying the tasks to be performed and assigning them to national climate-policy stakeholders according to their respective mandates and responsibilities.
- Propose measures to strengthen NCP implementation, facilitate access to international public and private financing, and promote foreign technological and financial partnerships.
- Identify opportunities to mobilize additional national resources to finance NCP implementation.

These objectives are aligned with Algeria's strategic documents, national development policy, and international commitments.

2.2. Principles and Preparation Methodology

2.2.1. Principles

The NCP was commissioned by the Government and covers the 2020-2025 period. It is one component of national climate policy. It was prepared by the National Climate Committee (CNC),⁵ chaired by the Minister of Environment and Renewable Energy (MEER). Its intended audiences are the Government, institutional sectors, socioeconomic operators, Algerian civil society, and international partners. Preparation of the NCP benefited from the support of Algerian-German Cooperation (GIZ) through national and international expertise.

The CNC defined the phases and methodology for preparing the NCP and verified compliance with them throughout the work. More specifically, the CNC formulated the working methodology adopted and approved by the sectors and institutions attending the national workshop that launched consultations for preparation of the NCP on 16 October 2017. The CNC also established the work schedule.

2.2.2. Preparation Stages

The NCP is the result of a comprehensive process comprising four major stages:

- Diagnosis:** The diagnosis identifies adaptation and mitigation priorities and needs. It is based on the current state of scientific knowledge concerning emissions by source and GHG removals by sinks, as derived from inventories. It also draws on completed and ongoing vulnerability analyses that identify and measure the impacts of climate change in Algeria.
- Strategy:** The strategy ensures a close link between climate-change mitigation and adaptation objectives, the seven pillars of the National Strategy for Environment and Sustainable Development (SNEDD), and Algeria's international commitments.
- Action plan:** The action plan presents the projects and actions identified by institutional sectors and civil society to address the established priorities and objectives.

⁵ In addition to MEER, the CNC brings together the institutional sectors most concerned by climate change: the Ministry of National Defense (MDN); Ministry of Foreign Affairs (MAE); Ministry of the Interior, Local Authorities, and Spatial Planning (MICLAT); Ministry of Water

Resources (MRE); Ministry of Energy (ME); Ministry of Industry and Mines (MIM); Ministry of Higher Education and Scientific Research (MESRS); Ministry of National Education (MEN); Ministry of Finance (MF); Ministry of Agriculture, Rural Development, and Fisheries (MADRP); Ministry of Housing, Urban Planning, and the City (MHUV); Ministry of Public Works and Transport (MTPT); Ministry of Health, Population, and Hospital Reform (MSPRH); Ministry of Commerce (MC); and National Economic and Social Council (CNES).

SOURCE PDF PAGE 19

2.2.2. Preparation Stages - Continued

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2.2.2. Preparation Stages - Continued

Implementation: This stage specifies the governance framework for organization, the means of carrying out the established activities, the timetable, monitoring, and evaluation.

Once the actions have been implemented, monitoring and a midterm evaluation of the NCP are required. On the basis of the evaluation's findings, advances in climate-change knowledge, and changes in the social, economic, and international contexts, the strategy, action plan, and implementation tools will be updated.

Figure 1 identifies the questions addressed at each stage. The proposed approach is cyclical and evolutionary.

FIGURE 1. STAGES IN THE PREPARATION OF THE NCP

NCP - Algeria

1. Diagnosis

What are the vulnerabilities to climate change?

Do they jeopardize achievement of the country's socioeconomic objectives?

What are the priorities?

2. Strategy

What are the NCP's objectives?

Are they consistent with the objectives of the SNEDD and with the country's international commitments?

3. Action Plans

What are the short- and medium-term action projects?

4. Implementation

What are the implementation tools, the NCP monitoring and evaluation system, and the financing mechanisms?

NCP Evaluation

2.2.3. Implementation Methodology

The approach adopted involved all institutional sectors and civil society from the outset. Although complex because of administrative constraints - numerous exchanges were required among sectors - and time constraints, the process had the advantage of providing guidance to the stakeholders concerned, raising their awareness of climate-change issues, assisting them in formulating their actions, and thereby strengthening their capacity in this field.

The CNC wanted the NCP to be prepared through the direct and close involvement of institutional sectors. The CNC considers that a participatory stakeholder approach ensures their involvement in implementing NCP actions. In other words, only the participation of sectors and potential stakeholders throughout the process of formulating and selecting actions can ensure the Plan's credibility, the conditions required for its implementation, and its consistency with national and sectoral strategies and Algeria's international commitments.

During their work, sector representatives, meeting in thematic groups, identified climate-related actions that had already been undertaken or were being planned in their respective fields.

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2.2.3. Implementation Methodology - Continued

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2.2.3. Implementation Methodology - Continued

The representatives then identified new actions whose implementation was considered necessary in light of the established priorities and objectives. The thematic-group work was guided by experts, who progressively prepared the guidance and tools required for each stage of NCP preparation.

Actions were identified and selected on the basis of the following criteria:

Sustainability: Definition of priorities; feasibility in terms of human and material resources; required timetable and durability of actions; and achievement of environmental and social co-benefits.

Effectiveness: Integration, to the greatest extent possible, of projects and actions into existing sectoral structures in order to avoid new organizational, human-resource, and material burdens.

Efficiency: Achievement of the established objectives at the lowest cost, so as to minimize impacts on the national economy, while ensuring strong cross-sectoral integration through intersectoral projects combining numerous actions.

Social cohesion: Consideration in particular of economic disparities and regional specificities.

International cooperation: Consistency with Algeria's climate objectives.

Transparency: Establishment of appropriate procedures so that actions are measurable, reportable, and verifiable through an MRV system.

Each proposed action is described using a standard template - the action sheet - which provides a systematic and uniform account of the content and characteristics of the identified action, including its objective, organization, lead institution and partners, coordination and implementation, indicators, and financing.

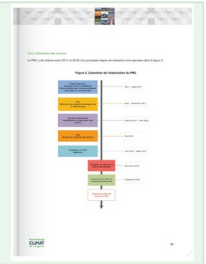
In total, more than 40 meetings were held and generated over 200 proposed actions. Following selection and consolidation, 155 actions were retained:

- 63 actions primarily concerning adaptation;
- 76 mitigation actions; and
- 16 actions addressing cross-cutting themes.

SOURCE PDF PAGE 21

2.2.4. Work Schedule

English translation below; original source-page thumbnail at right.



2.2.4. Work Schedule

The NCP was prepared between 2011 and 2018. The principal implementation stages are presented in Figure 2.

FIGURE 2. TIMETABLE FOR PREPARING THE NCP

- Diagnostic phase: vulnerability assessment and review of national and international actions and strategies - 2011 to July 2017
- CNC: definition of objectives and adoption of the methodology - August to September 2017
- Thematic groups: identification and description of actions - October 2017 to April 2018
- CNC: review and validation of actions - May 2018
- Finalization and drafting of the NCP - June to July 2018
- Validation of the NCP by the CNC and the sectors - December 2018
- Submission of the NCP to the Council of Government - September 2019
- Dissemination and implementation of the NCP

SOURCE PDF PAGE 22

3. CLIMATE CONTEXT

English translation below; original source-page thumbnail at right.

**3. CLIMATE CONTEXT**

This section is organized as follows:

The results of Algeria's latest GHG-emissions inventory are reviewed.

The characteristics of Algeria's climate are briefly presented, together with evidence of recent changes.

The consequences of climate change for Algeria's economy and society are briefly described on the basis of information available to date.

The international climate-change regime and Algeria's resulting commitments are presented.

Finally, the legislative framework and information on the costs of climate-change inaction and action are reviewed.

3.1. Greenhouse-Gas Emissions

The national inventory for the year 2000⁶ (MATE, 2010) indicates that total direct GHG emissions amounted to 117,310 Gg, or 117.31 million metric tons of CO₂ equivalent, in 2000. For the same year, CO₂ removal by forestry was estimated at 14.167 million metric tons of CO₂ equivalent, resulting in net emissions of 103.143 million metric tons of CO₂ equivalent. With the population estimated at 29,726,500 in 2000, gross emissions amounted to 3.95 metric tons of CO₂ equivalent per capita. For comparison, average global GHG emissions were 4.97 metric tons of CO₂ equivalent per capita in 2015.⁷

The precision of the estimates was assessed on the basis of expert judgment and the default values proposed in the IPCC good-practice guidance (GPG 2000 and 2006). For the inventory as a whole, the uncertainty margin was estimated at 12.90%. Algeria's total gross emissions therefore fall, within a 95% probability range, between 102.17 and 132.434 million metric tons of CO₂ equivalent. Net emissions fall between 89.831 and 116.44 million metric tons of CO₂ equivalent.

The energy sector is the largest emitter, accounting for 75% of total emissions. Emissions from this sector result from energy consumption (46%); hydrocarbon production, processing, and transport (20%); and natural-gas liquefaction (8%). Agriculture, land-use change, and forestry generate 11% of total GHG emissions. Waste and industrial processes account for 10% and 5% of total emissions, respectively. Of waste-sector emissions, 95% are emitted as methane from landfills; of industrial-process emissions, 60% consist of CO₂ from the cement industry.

3.2. Climate**3.2.1. Climate Zones**

From north to south, Algeria's climate changes markedly, passing from a humid Mediterranean climate to a dry desert environment through a semi-arid transitional zone. This climate reflects the combined influence of mid-latitude atmospheric circulation and tropical and Saharan circulation. The Mediterranean forms a transition zone between these climatic components.

Algeria's bioclimate can therefore be divided into the following zones:

A Mediterranean climate along the coast and in the Tell Atlas, with rare winter frosts and hot summers. The eastern part receives more rainfall than the western part, with up to 2,000 mm per year and snow-covered peaks from October to July.

An arid climate south of the Tell Atlas, dry and tropical in character and marked by a large winter day-night temperature range - 36°C during the day and 5°C at night. Autumn and spring rainfall is influenced by moist air advection from the coast.

⁶ The methodology used was the 1996 IPCC methodology and the 2000 Good Practice Guidance. In some cases, the 2006 IPCC guidance was used where required by the availability of unit emission factors or the methodology.

⁷ Data from the Carbon Dioxide Information Analysis Center, Environmental Sciences Division, Oak Ridge National Laboratory, Tennessee, United States, as cited in the source document through the World Bank indicator database:
<https://donnees.banquemondiale.org/indicateur/EN.ATM.CO2E.PC>

SOURCE PDF PAGE 23

3.2.1. Climate Zones - Continued

English translation below; original source-page thumbnail at right.



3.2.1. Climate Zones - Continued

Rainfall in the arid zone is more abundant in the west than in the east. The influence of the desert reaches the coast through the sirocco, a dry, hot wind that blows from south to north.

A continental climate on the High Plains and in the Saharan Atlas. Temperatures may fall below 0°C in winter and exceed 40°C in summer. Rainfall is scarce, particularly on the High Plains of Oran and Constantine.

A desert climate with rare and highly irregular rainfall, sometimes occurring as thunderstorms. The Sahara is one of the hottest regions in the world, where daytime temperatures may reach or exceed 50°C. Nights, however, are very cold, especially in winter, when freezing is frequent.

3.2.2. Temperature

Temperature patterns distinguish the different climatic regions: the coast, Tell Atlas, High Plains and Steppe, and Sahara. In winter, the High Plains and Steppe are colder than the Tell Atlas, the coast, and the Sahara. January is the coldest month of the year; mean temperature is 11.4°C at Ghazaouet and 10.0°C in Algiers.

In summer, mean temperature is 25.5°C in Algiers, 25.4°C at the Constantine station, and 24.0°C in Souk Ahras. In both summer and winter, the coast benefits from the moderating effect of the sea (MATE, 2010).

3.2.3. Water Resources and Rainfall

Algeria's conventional water-resource potential is estimated overall at 19.4 billion m³ per year, equivalent to 500 m³ per inhabitant per year. This places Algeria among countries poor in water resources, based on the World Bank scarcity threshold of 1,000 m³ per inhabitant per year.

According to MRE (2011), potential renewable water resources total 14.4 billion m³, 95% of which are located in the north of the country. This renewable-water potential consists of 11.4 billion m³ of surface water and 3 billion m³ of groundwater. Nonrenewable water resources amount to 5 billion m³ and are located entirely in the south.

Nonconventional water resources, estimated from the capacity of existing facilities, include desalination (840 million m³/year) and treated wastewater (400 million m³/year).

November and December are the rainiest period, particularly in the Tell Atlas, which receives more rain than the coast and the High Plateaus. In January, monthly rainfall is 3.7 mm at In Salah, 33.4 mm at Djelfa, 80.0 mm in Algiers, and 120.5 mm at Miliana. In summer, rain is rare and occurs more often along the coast than elsewhere. In autumn, rainfall patterns are nearly the same in the Tell Atlas and the High Plateaus. Annual rainfall is 15.8 mm at In Salah, 347.4 mm at Djelfa, 686.6 mm in Algiers, and 827.3 mm at Miliana (MATE, 2001).

3.2.4. Major Agroecological Regions

Algeria's climatic characteristics define five major agroecological regions, which determine the agricultural activities and production systems practiced.

a) Sahel and Coastal Areas

Because of particularly favorable climatic conditions, this agroecological region is occupied mainly by high-value crops: market gardening, especially greenhouse cultivation, and fruit growing in areas including Mostaganem, Algiers, Tipaza, Boumerdès, and Jijel.

The production system is generally intensive, while crop rotations extend over three, four, or even five years. Pesticide and fertilizer use is relatively high in greenhouse cultivation.

b) Sub-Coastal Plains

These agroecological areas are characterized by often heavy soils and a relatively favorable climate. Mixed farming is predominant - vegetable crops, forage crops, cereals, and fruit trees - together with cattle raising. Three-year rotations are most common, and the availability of water resources plays a decisive role in intensifying the cropping systems used.

SOURCE PDF PAGE 24

3.2.4. Major Agroecological Regions - Continued

English translation below; original source-page thumbnail at right.



3.2.4. Major Agroecological Regions - Continued

The sub-coastal plains also include a large part of the forest estate, represented by extensive, fairly densely wooded massifs, particularly in the 32 northern wilayas - from El Tarf to Tlemcen and, in a second belt, from Souk Ahras to Sidi Bel Abbès.

c) Inland Plains, High Cereal Plains, and High Plateaus

This region has several characteristics: cold to very cold winters and hot to very hot, dry summers. Because rainfall is relatively low, water availability is a decisive factor in selecting cropping systems:

On the plains, open-field vegetable crops in the main and late seasons, forage crops, cereals, and fruit growing are practiced where irrigation water is available. Where it is not, a two-year cereal-fallow system is generally used, with a cereal-forage-fallow rotation.

On the high plains and in some High Plateau areas, cereal growing is combined with livestock raising, which provides crop-livestock farmers with greater security against increasingly recurrent climatic shocks.

d) Steppe High Plains

Located between the 400 mm isohyet in the north and the 150 mm isohyet in the south, this region extends across 32 million hectares from east to west.

The ancestral and dominant practice is pastoral raising of sheep, goats, and camels, together with occasional cereal cultivation - mainly barley - on flood-spreading areas, used primarily as supplementary livestock feed.

Marking the southern limit of the forested area, the latter two regions are characterized by less dense and less wooded massifs with relatively limited species diversity, but covering large territories, such as those of the Saharan Atlas in Laghouat-Djelfa and the Aurès in Batna-Khenchela.

e) Saharan and Oasis Areas

Over the centuries, the scarcity of arable land and water, together with the climate, has led to the development of a highly complex multi-tier cultivation system: date palms, fruit trees, vegetables, cereals, and forage crops.

3.3. Recent Climate Trends

Analysis of recent climate trends shows that the effects of climate change are now perceptible in the Mediterranean region and are increasingly causing recurrent climate anomalies and extreme events (Oualkacha et al., 2017; Giorgi et al., 2008).

The scientific community considers these changes unavoidable. In other words, current and projected global mitigation efforts can reduce them only partially. Algeria's economy and society must therefore adapt to climate change in order to lessen its adverse consequences.

In the Maghreb and, more specifically, Algeria, recent climate trends show warming greater than the global average.

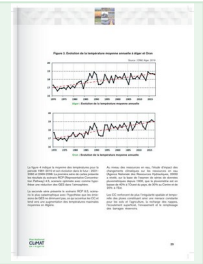
While the worldwide rise in temperature during the twentieth century was 0.74°C, the rise in the Maghreb ranged from 1.5°C to 2°C depending on the region - more than twice the global average increase. The decline in precipitation ranges from 10% to 20% (Tabet-Aoul, 2008).

Figure 3 shows temperature trends at the Algiers and Oran stations from 1970 to 2017. Based on daily temperature data, it indicates that temperature rose by more than 1.5°C during this period.

SOURCE PDF PAGE 25

FIGURE 3. TREND IN MEAN ANNUAL TEMPERATURE IN ALGIERS AND ORAN

English translation below; original source-page thumbnail at right.

**FIGURE 3. TREND IN MEAN ANNUAL TEMPERATURE IN ALGIERS AND ORAN**

Source: ONM, Algiers, 2019

Upper graph: Algiers - Trend in mean annual temperature, 1970-2017.

Lower graph: Oran - Trend in mean annual temperature, 1970-2017.

The vertical scale ranges from approximately 16°C to 20°C. The black line shows annual mean temperature and the dashed trend line rises over time in both cities.

Figure 4 presents mean temperatures for the 1981-2010 period and their projected future change in 2031-2060 and 2069-2098. The first series of maps shows results for Representative Concentration Pathway (RCP) 4.5, an optimistic scenario based on the assumption of reduced GHG concentrations in the atmosphere.

The second series presents RCP 8.5, the most severe scenario, which assumes that GHG emissions do not decline. This intensifies climate change and leads toward an increase in mean maximum temperatures in Algeria.

Regarding water resources, the study of climate-change impacts on water resources (National Water Resources Agency, 2009), based on analysis of rainfall-data series dating from 1900, found that rainfall had declined by 40% in the west of the country, 30% in the center, and 20% in the east.

Climate change also increases the spatial and temporal irregularity of rainfall, thereby posing a continuing threat to soils and agriculture, groundwater recharge, surface runoff, reservoir siltation, and reservoir filling.

SOURCE PDF PAGE 26

As an example, Figure 5 shows the trend in precipitation at the Oran station between 1926 and 2006 and indicates a decline of approximately 15%.

English translation below; original source-page thumbnail at right.



As an example, Figure 5 shows the trend in precipitation at the Oran station between 1926 and 2006 and indicates a decline of approximately 15%.

Examination of maps prepared by ANRH for 1942-1989 and 1965-2004 (Figure 6) shows that the 300 mm isohyets shifted significantly toward the north of the country (ANRH, 2009), constituting one indicator of climate change in Algeria.

Figure 7 illustrates mean annual precipitation during the wet period - when notable rainy episodes occur between September and May - for 1981-2010, and the projected change in this mean for 2031-2060 and 2069-2098. The two map series are based on two IPCC scenarios: RCP 4.5 on top and RCP 8.5 on the bottom.

FIGURE 4. MAP OF MEAN MAXIMUM TEMPERATURE DURING THE RAINY SEASON BETWEEN SEPTEMBER AND MAY

Source: MEER & GIZ, 2017

Map panels:

Top row: RCP 4.5 - 1981-2010 baseline; projected change for 2031-2060; projected change for 2069-2098.

Bottom row: RCP 8.5 - 1981-2010 baseline; projected change for 2031-2060; projected change for 2069-2098.

Location: Algeria.

Description in the figure: The maps illustrate mean maximum temperatures during the rainy season for the 1981-2010 period and the projected change in this mean for 2031-2060 and 2069-2098. The first map series presents RCP 4.5 results; the second presents RCP 8.5 results. [The source graphic's parenthetical list of months appears inconsistent with the caption; it is retained in the original-page image rather than silently corrected.]

Data sources: Bias-corrected CORDEX-Africa ensemble; eight simulations generated from four GCMs - CNRM-CM5, EC-EARTH, HadGEM2-ES, and MPI-ESM-LR - and two RCMs - SMHI-RCA4 and CLMcom-CCLM4-8-17 - run with the same GCMs.

Legend:

Mean maximum temperature (°C)

Change in mean maximum temperature (°C)

Isolines (°C)

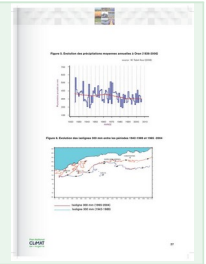
Scale: 1:15,000,000, A3 format

Reference in the figure: The map was produced under the National Climate Plan Support Project (APNC), within the framework of cooperation between Germany and Algeria.

SOURCE PDF PAGE 27

FIGURE 5. TREND IN MEAN ANNUAL PRECIPITATION IN ORAN, 1926-2006

English translation below; original source-page thumbnail at right.

**FIGURE 5. TREND IN MEAN ANNUAL PRECIPITATION IN ORAN, 1926-2006**

Source: M. Tabet-Aoul (2008)

Graph labels:

Vertical axis: Annual rainfall in millimeters

Horizontal axis: Year

The annual series and dashed trend line show a long-term decline.

FIGURE 6. SHIFT IN THE 300 MM ISOHYETS BETWEEN 1942-1989 AND 1965-2004

Map labels include Tlemcen, Saïda, Djelfa, Bordj Bou Arreridj, and Constantine.

Legend:

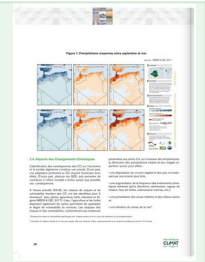
Red dotted line: 300 mm isohyet, 1965-2004

Blue dotted line: 300 mm isohyet, 1942-1989

SOURCE PDF PAGE 28

FIGURE 7. MEAN PRECIPITATION BETWEEN SEPTEMBER AND MAY

English translation below; original source-page thumbnail at right.

**FIGURE 7. MEAN PRECIPITATION BETWEEN SEPTEMBER AND MAY**

Source: MEER & GIZ, 2017

Map panels:

Top row: RCP 4.5 - 1981-2010 baseline; projected change for 2031-2060; projected change for 2069-2098.

Bottom row: RCP 8.5 - 1981-2010 baseline; projected change for 2031-2060; projected change for 2069-2098.

Location: Algeria.

Description: The maps illustrate mean precipitation during the period with notable rainy episodes - between September and May - for 1981-2010 and the projected change in this mean for 2031-2060 and 2069-2098. The first map series presents RCP 4.5 results; the second presents RCP 8.5 results.

Data sources: Bias-corrected CORDEX-Africa ensemble; eight simulations generated from four GCMs - CNRM-CM5, EC-EARTH, HadGEM2-ES, and MPI-ESM-LR - and two RCMs - SMHI-RCA4 and CLMcom-CCLM4-8-17 - run with the same GCMs.

Legend:

Mean precipitation (mm)

Change in precipitation (mm)

400 mm threshold significant for agricultural activities

Isohyets (mm)

Scale: 1:15,000,000, A3 format

Reference in the figure: The map was produced under the National Climate Plan Support Project (APNC), within the framework of cooperation between Germany and Algeria.

3.4. Impacts of Climate Change

Identifying the consequences of climate change for Algeria's economy and society is a priority. Relevant adaptation requires anticipating these effects. At the same time, mitigation of GHG emissions should contribute to the global effort to limit these consequences as much as possible.

As of 2018, climate-change risk and vulnerability chains had been identified for five sectors in Algeria - water, fisheries, agriculture, forests, and industry (MEER & GIZ, 2017). Water, agriculture, and forests also have maps that show the spatial distribution of territorial vulnerability. In accordance with the evidence presented above concerning rising temperatures, declining total precipitation, and its uneven distribution, risk and vulnerability analyses indicate the following effects:

Degradation of vegetation cover and soils, resulting in greater erosion;

Increased frequency of extreme climate events, including torrential rain, drought, heat waves, forest fires, and coastal flooding;

Disruption of coastal areas and marine environments; and

Sea-level rise.⁹

⁸ Sector-specific risk and vulnerability analyses are being conducted or planned.

⁹ Sea-level rise in Algeria appears to be three to four times faster, as measured at Sfax, than the global average of 1.5-2 mm per year.

SOURCE PDF PAGE 29

Figure 8 illustrates the chain of climate-change consequences. A substantial temperature increase, combined with a significant decline in precipitation and increased evaporation, w

English translation below; original source-page thumbnail at right.

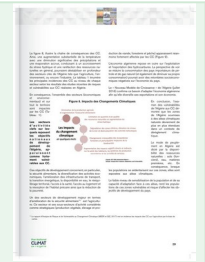


Figure 8 illustrates the chain of climate-change consequences. A substantial temperature increase, combined with a significant decline in precipitation and increased evaporation, would increase water stress and reduce natural resources generally. This could profoundly destabilize key Algerian sectors such as agriculture, the environment, and industry. Table 1 lists the principal climate-change effects in each sector, based on recent climate-risk and vulnerability studies conducted in Algeria.

Consequently, every economic and environmental sector across the entire country is affected by climate change (Table 1).

The key sectors on which Algeria's national development objectives depend appear highly vulnerable to climate change.

These development objectives concern, in particular, food security; diversification of economic activities; improvement of transport infrastructure; the energy transition; water availability; territorial rebalancing; access to health care; access to housing and the elimination of precarious housing; and poverty reduction.

Agriculture is a major development sector for improving food security.¹⁰ This sector and its strategic subsectors - crop production, livestock and meat production, forestry, and fisheries - nevertheless appear heavily affected by climate change (Figure 9).

Algeria's economy also relies on hydrocarbon extraction and exports. The prospect of reduced oil and natural-gas consumption in importing countries - and of reducing Algeria's own consumption - could have adverse socioeconomic effects on the national economy.

Algeria's "New Growth Model" of July 2016 confirms the need to adapt the Algerian economy so that it diversifies both its exports and its economic base.

In conclusion, examination of Algeria's climate vulnerabilities shows that the parts of the country exposed to natural climate hazards are becoming increasingly extensive in a context of climate disruption.

Patterns of settlement in Algeria are determined by the availability of natural resources - land and soils, water, raw materials, and others. When populations settle permanently in such areas, they are exposed to climate hazards.

Low public awareness and limited adaptive capacity in the face of these hazards make populations in these areas vulnerable and may affect the country's development objectives.

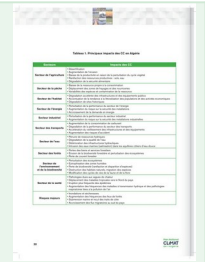
FIGURE 8. IMPACTS OF CLIMATE CHANGE - IN BRIEF

- Decline in agricultural production and increase in food insecurity;
- Quantitative and qualitative limitations on natural resources and increased water stress;
- Degradation of coastal areas, sea-level rise, and disruption of fishing activities;
- Irreversible changes to forest ecosystems and major disruptions to biodiversity; and
- Increased direct and indirect adverse impacts on human health, secondary production systems, and service activities.

Source in the figure: Adapted from the IPCC Fifth Assessment Report, Climate Change 2014: Impacts, Adaptation, and Vulnerability.

¹⁰ The Climate-Change Risk and Vulnerability Analysis report (MEER & GIZ, 2017) demonstrates the impacts of climate change on agriculture through maps.

SOURCE PDF PAGE 30

TABLE 1. MAIN IMPACTS OF CLIMATE CHANGE IN ALGERIA*English translation below; original source-page thumbnail at right.***TABLE 1. MAIN IMPACTS OF CLIMATE CHANGE IN ALGERIA****Agriculture**

- Desertification
- Increased erosion
- Lower productivity due to disruption of the plant-growth cycle
- Increasing scarcity of productive resources: soil and water
- Deterioration of food security

Fisheries

- Decline in resources available for consumption
- Shifts in spawning and nursery areas
- Changes in species composition and contamination of resources

Housing

- Accelerated deterioration of infrastructure and public facilities
- Intensification of the tendency for populations and economic activities to concentrate along the coast
- Degradation of historic sites

Energy

- Disruption of energy-sector performance
- Increased safety risks at facilities
- Increased energy demand

Industry

- Disruption of industrial-sector performance
- Increased safety risks at industrial facilities

Transport

- Increased fuel consumption
- Deterioration of transport-sector performance
- Accelerated aging of infrastructure and equipment
- Increased accident risk

Water

- Shortage of water resources
- Deterioration of water quality
- Damage to hydraulic infrastructure
- Seawater intrusion and salinization in freshwater coastal aquifers

Forests

- Loss of forest goods and services
- Erosion of forest biodiversity and disruption of ecosystems
- Loss of forest cover

Environment and Biodiversity

- Disruption of ecosystems

Eutrophication of wetlands

Biodiversity loss, including increasing scarcity and disappearance of species

Destruction of natural habitats and migration of species

Changes in the life cycles of fauna and flora

Health

Illnesses caused by heat waves

Northward movement of tropical diseases

More frequent outbreaks of epidemics

Increased frequency of waterborne diseases and respiratory illnesses related to air pollution

Major Risks

Floods and droughts

Increased frequency of forest fires

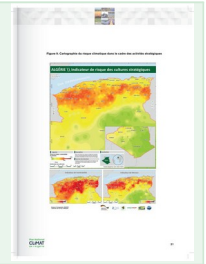
Coastal flooding and shoreline retreat

Increased migration flows in the south of the country

SOURCE PDF PAGE 31

FIGURE 9. MAPPING OF CLIMATE RISK IN CONNECTION WITH STRATEGIC ACTIVITIES

English translation below; original source-page thumbnail at right.

**FIGURE 9. MAPPING OF CLIMATE RISK IN CONNECTION WITH STRATEGIC ACTIVITIES**

Main map title: ALGERIA - Strategic Crops Risk Indicator

Legend: Level of risk, vulnerability, and threat, from 0% to 100%; the scale marks 60%. Wilaya boundaries are shown.

Description: The map illustrates the level of risk that the population's basic food security may not be ensured with respect to cereals and milk through forage resources.

Data sources: Composite indicator calculated from vulnerability and threat indicators developed at the municipal level.

Location: Algeria.

Lower maps:

Vulnerability Indicator

Threat Indicator

Analysis and cartography: BNEDER, 28 February 2017; GIZ / BNEDER.

SOURCE PDF PAGE 32

3.5. International Climate-Change Regime

English translation below; original source-page thumbnail at right.

**3.5. International Climate-Change Regime****3.5.1. Multilateral Environmental Agreements**

Algeria signed and ratified the following United Nations environmental conventions: (i) the United Nations Framework Convention on Climate Change (UNFCCC/CCNUCC) in 1993; (ii) the United Nations Convention on Biological Diversity (UNCBD) in 1995; and (iii) the United Nations Convention to Combat Desertification (UNCCD) in 1996.

The ultimate objective of the United Nations Framework Convention on Climate Change¹¹ is to stabilize GHG concentrations in the atmosphere at a level that prevents dangerous anthropogenic interference with the climate system. States commit to taking measures to combat climate change and its consequences, including national GHG inventories; programs to mitigate and adapt to change; and the application and dissemination of appropriate technologies. The Rio Earth Summit also established the Global Environment Facility (GEF) to address environmental problems at the global level while encouraging sustainable development nationally. The GEF provides financial support for projects developed in developing and transition countries in the areas of biodiversity, climate change, international waters, land degradation, the ozone layer, and persistent organic pollutants.

3.5.2. Kyoto Protocol

The Kyoto Protocol (KP, 1997) extends the United Nations Framework Convention on Climate Change and pursues the objective of reducing GHG emissions.¹² Thirty-seven industrialized countries listed in Annex I committed to reducing GHG emissions. Developing countries not listed in Annex I were exempted from quantified emission-reduction commitments under the Protocol.

The Kyoto Protocol established three flexibility mechanisms for reducing GHG emissions. First, international emissions trading concerns the carbon market through the exchange of carbon credits. Second, there are two project mechanisms: Joint Implementation and the Clean Development Mechanism. Joint Implementation allows a company to invest in projects that reduce or avoid GHG emissions by another company in the same country or in another Annex I country in order to obtain emission credits.

The Clean Development Mechanism enables developed Annex I countries to meet their reduction targets by investing in GHG-emission-reduction projects in developing or emerging non-Annex I countries.

3.5.3. Paris Agreement

During negotiations on continuation of the Kyoto Protocol beginning in 2007, industrialized and developing countries disagreed about developing-country participation in emission-mitigation efforts. Industrialized countries made continuation of their mitigation efforts conditional on developing-country participation in global efforts.

Developing countries asserted their right to development and recalled the historical responsibility of developed countries for global warming. They therefore called for observance of the Framework Convention principle that responsibilities are common but differentiated according to countries' respective capabilities. They nevertheless agreed to act voluntarily, provided developed countries honored their commitments concerning finance, technology transfer, and capacity-building.

On 12 December 2015, the Paris Agreement was adopted by COP21.¹³ By June 2017, 178 Parties - 177 countries and the European Union - had ratified it, including Algeria in October 2016. The Paris Agreement is an additional protocol to the United Nations Framework Convention on Climate Change.

The Agreement is legally binding but does not provide sanctions for countries that fail to comply. Its objective continues on the next source page.

¹³ The Agreement entered into force on 4 November 2016.

¹⁴ When formulating their NDCs, countries set mitigation and adaptation objectives. They first formulated Intended Nationally Determined Contributions following decisions at the Lima summit (COP20, 2014) in preparation for COP21 in Paris in 2015. After adoption of the Paris Agreement, all Parties' communicated INDCs - unless updated after the first submission - automatically became the countries' first NDCs.

[Footnotes 11 and 12 are referenced in the source text, but their footnote text is not printed on this page of the source PDF. No wording has been supplied here by inference.]

SOURCE PDF PAGE 33

3.5.3. Paris Agreement - Continued

English translation below; original source-page thumbnail at right.



3.5.3. Paris Agreement - Continued

The objective of the Paris Agreement is to keep global warming below 2°C by 2100 relative to preindustrial levels. The Agreement requires each country to communicate a Nationally Determined Contribution (NDC) every five years. A global stocktake is to be carried out every five years, with the first planned for 2023. The stocktakes are intended to lead to strengthened NDCs in accordance with the principle of progression set out in the Agreement.¹⁵ The Agreement states that developed countries must assist developing countries in mitigation and adaptation through finance and technology transfer.

Algeria submitted its NDC, in which it undertakes to reduce its GHG emissions relative to the reference scenario by 7% by 2030 on a voluntary basis using national resources, and by 22% conditional on international support.

3.6. Legislative Framework

The Law on Environmental Protection within the Framework of Sustainable Development and the Law on the Promotion of Renewable Energy refer directly to climate change.

The Law on Environmental Protection within the Framework of Sustainable Development classifies GHGs as air pollutants (Article 4, Title II) and provides for the possibility of financial and customs incentives for companies importing equipment that eliminates or reduces GHG emissions or any other form of air pollution (Title V, Article 76). The Law on the Promotion of Renewable Energy states that renewable energy is a means of reducing GHG emissions (see Articles 2, 3, 6, 13, and 17).

Legislation indirectly related to climate change concerns energy, water, agriculture, biodiversity, spatial planning, cities, and the coast. Table 2 lists legal texts relevant to climate change.

The range of existing legal instruments therefore provides a solid foundation for establishing Algeria's climate policy and developing its NCP.

TABLE 2. LEGISLATIVE FRAMEWORK

2012 - Law No. 12-07 establishing the Wilaya Code

2011 - Law No. 11-10 concerning municipalities

2011 - Law No. 11-02 concerning protected areas within the framework of sustainable development

2010 - Law No. 10-02 approving the SNAT 2030

2008 - Law No. 08-16 establishing agricultural policy guidelines

2007 - Law No. 07-06 concerning the management, protection, and development of green spaces

2006 - Law No. 06-06 establishing city-policy guidelines

2005 - Law No. 05-12 concerning water

2004 - Law No. 04-09 concerning the promotion of renewable energy within the framework of sustainable development

2004 - Law No. 04-03 concerning the protection of mountain areas within the framework of sustainable development

2004 - Law No. 04-20 concerning the prevention of major risks and disaster management within the framework of sustainable development

1990, 2004 - Law No. 04-05 concerning development and urban planning

2003 - Law No. 03-10 concerning environmental protection within the framework of sustainable development

2002 - Law No. 02-01 concerning electricity and gas distribution

2002 - Law No. 02-02 concerning protection and enhancement of the coastal zone

2002 - Law No. 02-08 concerning the conditions for creating and developing new cities

2001 - Law No. 01-20 concerning spatial planning and sustainable territorial development

2001 - Law No. 01-11 concerning fisheries and aquaculture

2001 - Law No. 01-19 concerning waste management, control, and disposal

1999 - Law No. 99-09 concerning energy management

1984, 1991 - Law No. 91-20 establishing the general forestry regime

¹⁵ The Agreement also provides a right of withdrawal by notification beginning three years after its entry into force.

SOURCE PDF PAGE 34

3.7. Economic Issues

English translation below; original source-page thumbnail at right.

**3.7. Economic Issues**

Economic considerations play an important role in preparing climate strategies. Assessing the costs of climate action and inaction is important both strategically and for management.

On the one hand, climate-change consequences affecting economic activities and the population may generate direct and indirect economic losses; this is the cost of inaction.

On the other hand, mitigation and adaptation measures intended to avoid climate change or minimize its impacts require financial resources; this is the cost of action. Despite the growing number of studies and case analyses, currently available evidence on the costs of climate action and inaction remains limited. It also generates debate about the degree of accuracy and relevance of such estimates.¹⁶

3.7.1. Cost of Inaction

From an economic perspective, priority should be given to actions that avoid a high cost of inaction relative to the cost of action. Available international studies tend to show that the policy costs of adaptation and mitigation account for only a small proportion of the potential cost of climate-change damage.

According to the Stern Review (Stern, 2006), the costs of inaction are substantially higher than the costs of action: investing 1% of global GDP each year in reducing GHG emissions could avoid an annual economic loss equivalent to between 5% and 20% of global GDP. The IPCC¹⁷ confirms these findings: the longer governments delay action, the heavier the burden on future generations. Doing nothing could reduce global GDP by 1% to 4% in 2030, 2% to 6% in 2050, and 3% to 12% in 2100.

As part of formulation of the SNEDD and PNAEDD, MEER (2018) estimated the costs of environmental degradation in Algeria, including degradation resulting from climate inaction. Using internationally recognized methodologies,¹⁸ Algeria's annual climate-change costs are estimated at between 181 billion Algerian dinars - 1.10% of GDP - and 303 billion dinars - 1.83% of GDP.

3.7.2. Cost of Action

Within the NCP process, sectors attempted to estimate systematically the cost of proposed mitigation and adaptation actions. For most proposed projects, however, this assessment proved difficult and remains incomplete, despite its major importance for identifying funders and implementation. At this stage, costs are reported only for actions that have already been budgeted.

Numerous international studies seek to estimate the costs of GHG-reduction efforts: that is, the opportunity cost of measures, actions, or concrete projects that avoid one metric ton of CO₂ emissions. This cost varies by emission source and therefore by country, economic sector, and company.¹⁹

This approach seeks to establish the "technical" value of carbon and indicate in which sector, country, and conditions reducing GHG emissions appears appropriate. A principal source of information on GHG-mitigation costs is macroeconomic analysis estimating the cost to an economy of achieving a given emission-reduction target.

¹⁶ Cost estimates must account for the fact that mitigation measures also address other pollutants. Reducing fossil-energy consumption to mitigate GHG emissions, for example, also reduces local air pollutants. These additional benefits of climate policies are called co-benefits and should be included in a complete assessment of the economic costs and benefits of climate policy (for example, Blanco et al., 2014; Smith et al., 2014). The same applies to adaptation. Strengthening early-warning capacity for natural disasters may improve public safety even without climate change. Adaptation and mitigation also generate new economic activities and thereby contribute to development of the green economy. Transition to a less GHG-intensive economy offers new opportunities for growth and innovation, including in renewable energy, heat and material recovery, and ecotourism.

¹⁷ See the IPCC reports reviewing economic research on climate change, published in 2007 and 2014.

¹⁸ The estimate is based on the average of two possible values per metric ton of CO₂. The high value - USD 30/tCO₂, or DZD 3,014/t - reflects the social value of carbon based on the cost of climate-change consequences for 2015 and is taken from the World Bank (2014). The low value - USD 4.6/tCO₂, or DZD 462/t - is based on the average price per metric ton of CO₂ in 2015-2016 on the European Union Emissions Trading System in 2015. An uncertainty margin of ±25% is applied to the selected average.

¹⁹ Assumed and observed differences among countries and sectors motivate emission-reduction trading systems, which allow emissions to be reduced where the cost is lowest.

SOURCE PDF PAGE 35

3.7.2. Cost of Action - Continued

English translation below; original source-page thumbnail at right.



3.7.2. Cost of Action - Continued

No macroeconomic study of this kind has been conducted for Algeria, with the exception of the renewable-energy field (Akbi et al., 2016).

Marginal GHG-emission-abatement cost curves constitute a second source of information on CO₂-emission-reduction costs (see McKinsey & Company, 2009). This source adopts a different, purely technical perspective that identifies the costs of reduction opportunities in an economy - only measures costing less than €100 per metric ton of CO₂ are included - and estimates the potential GHG reduction for each option considered. No such curve has been developed for Algeria either.

SOURCE PDF PAGE 36

4. STRATEGIC ANCHORING

English translation below; original source-page thumbnail at right.

**4. STRATEGIC ANCHORING**

The NCP was designed to be consistent with Algeria's economic and social strategy, including the new economic growth model; the Government Action Plan; the National Spatial Planning Scheme (SNAT 2017); environment-related strategies - the National Strategy for Environment and Sustainable Development and the National Biodiversity Strategy - and Algeria's international obligations.

4.1. Strategic Planning Documents

The National Strategy for Environment and Sustainable Development (SNEDD) for 2017-2035 is intended to become the strategic reference document for the environment. In its current version, Pillar 5 is devoted to climate change. It explicitly mentions the NCP - "institutionalize, update, and implement the National Climate Plan and ensure its consistency with the Paris Agreement" - and the NDC - "implement the Nationally Determined Contribution and ensure that it is updated." Its objectives include GHG mitigation, climate-change adaptation, and stronger climate planning. The NCP must ensure consistency with national strategic documents and contribute to meeting their climate objectives.

The new economic model for 2020-2035 does not explicitly mention climate change. Its fifth pillar, however, addresses the energy transition and therefore directly supports mitigation through improved energy efficiency and renewable-energy development. The model establishes the national renewable-energy and energy-efficiency development program, which seeks to reach 22,000 megawatts of renewable capacity in Algeria's energy mix by 2030, as stated in the NDC.

The National Spatial Planning Scheme (SNAT) treats climate risks as major risks and helps identify their consequences. It indicates priority climate-adaptation needs across the national territory. Developed within the SNAT framework, the National Biodiversity Strategy and Action Plan (SPANB) seeks to conserve, protect, restore, and enhance Algeria's natural ecosystems. The SPANB highlights climate-change impacts on biodiversity and identifies means of ensuring Algeria's climate resilience. Synergy with the NCP is considered essential.

Climate change is addressed in SPANB Objectives 15 and 17:

Objective 15: Reduce risks affecting biodiversity, particularly those related to various forms of pollution, climate-change impacts, invasive alien species, and other poor management practices.

Objective 17: Integrate climate-adaptation and natural-risk and disaster-prevention approaches into ecosystem management.

The Government Action Plan explicitly mentions climate change as a factor that intensifies the need to increase water availability over the medium and long terms in order to ensure food security. It highlights one of the major constraints climate change creates in Algeria and directly reinforces the introduction of adaptation measures.

4.2. International Commitments

Algeria's commitments arise from ratification of the three United Nations environmental conventions described in Section 3.5.1. They are reflected specifically in the NDC under the Paris Agreement and in the United Nations 2030 Sustainable Development Goals.

For mitigation, the NDC states that by 2030 Algeria commits to a voluntary 7% reduction using national resources and a conditional 22% reduction with international support. The energy transition is also specified. This reflects the political commitment expressed by the Council of Ministers in May 2017, which made implementation of the national renewable-energy and energy-efficiency program to 2030 a priority.

SOURCE PDF PAGE 37

4.2. International Commitments - Continued

English translation below; original source-page thumbnail at right.

**4.2. International Commitments - Continued**

The Algerian Government intends to undertake structural reforms and public policies related to these international commitments. The reforms include revision of the Hydrocarbons Law to implement a new Sonatrach strategy, SH2030, incorporating renewable energy into oil and gas fields. Another law concerning local authorities, currently being prepared, provides for the integration of energy efficiency and renewable-energy development. For the same purpose, the 1999 law on energy management is also under revision.

For adaptation, the NDC states the following objectives:

Establish monitoring and early-warning systems for major climate risks, including floods, coastal inundation, forest fires, and drought.

Strengthen ecosystem resilience to minimize the effects of major climate risks.

Protect public health against the resurgence of tropical diseases, epidemics, and heat waves.

Combat erosion and restore degraded land as part of efforts to combat desertification.

Integrate climate-change effects into sectoral strategies, particularly agriculture, water resources, human health, public works, and transport.

Integrate climate-change effects in order to ensure food security and national security.

The NDC specifies that achieving these objectives requires diversified international support, including finance, capacity-building, and technology transfer.

TABLE 3. NDC OBJECTIVE AND FLAGSHIP MITIGATION ACTIONS

NDC mitigation objective:

Achieve an energy transition and economic diversification in order to meet Algeria's sustainable-development objectives.

Flagship actions stated in the NDC

Energy Sector - Reference to the National Renewable-Energy and Energy-Efficiency Development Program

Increase national electricity generation from renewable energy by 2030.

Generalize the use of efficient lighting.

Improve the thermal insulation of housing.

Increase the shares of liquefied petroleum gas and natural gas in fuel consumption.

Reduce the volume of gas flared.

Waste Sector

Recover value from waste.

Compost organic and green waste.

Capture methane from engineered landfill centers and wastewater-treatment plants for beneficial use and energy recovery.

Forestry Sector

Afforestation and reforestation.

Prevent forest fires and improve firefighting capacity.

Awareness, Information, and Education

Provide information, awareness-raising, and communication on climate-change issues and challenges.

Establish a national climate-change education, training, and research program.

SOURCE PDF PAGE 38

Table 3 specifies the means recommended by the NDC for achieving mitigation and adaptation objectives and identifies the action programs to be implemented.

English translation below; original source-page thumbnail at right.



Table 3 specifies the means recommended by the NDC for achieving mitigation and adaptation objectives and identifies the action programs to be implemented.

For formulation of the NCP, the NDC served as the framework for work by the thematic groups. This made it possible to specify and detail implementation arrangements for the planned actions and to propose additional actions considered necessary to achieve the objectives.

Among the United Nations Sustainable Development Goals, Goal 13²⁰ explicitly calls for urgent action to combat climate change and its impacts.

²⁰ SDG 13 is set out as follows: 13.1 Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries. 13.2 Integrate climate-change measures into national policies, strategies, and planning. 13.3 Improve education, awareness-raising, and human and institutional capacity on climate-change mitigation, adaptation, impact reduction, and early warning. 13.a Implement the commitment undertaken by developed-country Parties to the United Nations Framework Convention on Climate Change to mobilize jointly USD 100 billion annually by 2020 from all sources to address the needs of developing countries in the context of meaningful mitigation actions and transparency on implementation, and make the Green Climate Fund fully operational by capitalizing it as soon as possible. 13.b Promote mechanisms for raising capacity for effective climate-change-related planning and management in least developed countries and small island developing States, including a focus on women, youth, and local and marginalized communities.

SOURCE PDF PAGE 39

5. PHASING AND ACTION PLAN

English translation below; original source-page thumbnail at right.

**5. PHASING AND ACTION PLAN****5.1. NCP Phases**

At the national level, the NCP timetable must be consistent with the periods covered by other national planning instruments.

The NCP is also incorporated into the PNAEDD five-year plan and defines projects and actions to be implemented over the short and medium terms. It therefore includes actions covering 2020-2035.

The adaptation component includes short- and medium-term actions. The mitigation and cross-cutting components include only short-term actions. This difference reflects the differing nature of adaptation, mitigation, and cross-cutting actions.

The adaptation component faces particular obstacles resulting from gaps in knowledge and in the availability of human and financial resources. Actions affected by these constraints are therefore planned for the medium term.

The mitigation component seeks to achieve the NDC objectives. Sectors have already prepared action programs that can be implemented in the short term.

The cross-cutting component seeks to establish the initial conditions - capacity-building, research, and communication - required to implement the NCP. It therefore concerns the short term exclusively.

The NCP is to be updated in 2023 for the 2025-2030 period on the basis of:

- Results of NCP action projects for 2020-2025;
- Changes in Algeria's economic, social, and environmental contexts;
- Outcomes of multilateral climate negotiations; and
- A possible update of the NDC in 2025.

At the international level, preparation of the 2023 global stocktake under the Paris Agreement entails:

Preparation of a roadmap for achieving the NDC's voluntary scenario, which provides for a 7% reduction in GHG emissions relative to a reference scenario by 2030. The roadmap must include all planned activities to mitigate GHG emissions in Algeria, emission-estimation methodologies, and methods for measurement, reporting, and verification. Actions by the various sectors that will be carried out using Algeria's own resources are included in this NCP. The list remains open so that any new action affecting GHG mitigation within the national territory may be added.

Development of a roadmap for the conditional scenario, which provides for a 22% reduction in GHG emissions relative to a reference scenario by 2030. This reduction requires international financial support, technology and know-how transfer, and strengthening of national capacities. With the participation of public and private sectors concerned with mitigation, Algeria must analyze its needs and identify all potential actions capable of mitigating GHG emissions by 2030. This stage should produce a cross-sectoral project portfolio identifying potential funders and eligibility conditions for international finance. International finance requires review and adaptation of Algerian regulatory texts - particularly to meet climate-finance eligibility criteria - to remove barriers to receiving international funds and implementing internationally supported projects in Algeria.

SOURCE PDF PAGE 40

5.1. NCP Phases - Continued

English translation below; original source-page thumbnail at right.



5.1. NCP Phases - Continued

The evolving nature of the NCP should be emphasized. Additional initiatives may be introduced during implementation according to Algeria's socioeconomic progress, available budgets, and technological innovation.

The action plan has three components:

- Socioeconomic adaptation of Algeria to climate change;
- Mitigation of GHG emissions; and
- Cross-cutting actions concerning climate governance, finance, awareness-raising, communication, training, capacity-building, and research and development.

5.2. Adaptation

Although not all effects of climate change are adverse and some create new opportunities, most climate-driven changes in the physical environment or biotopes are negative and threaten the sustainability, composition, resilience, or productivity of natural and managed ecosystems (MEER & GIZ, 2017).

Adaptation is the process of adjustment in natural or human systems in response to actual or expected climatic stimuli or their effects, in order to moderate harm or exploit beneficial opportunities (IPCC, 2014).

Adaptation is an evolving process, not an outcome (GIZ/WRI, 2011). Several types of adaptation are distinguished: anticipatory or reactive, private or public, and autonomous or planned. Examples include building dikes along rivers or coasts, adopting new seed varieties, adapting production calendars, and increasing storage capacity.

Within the NCP, adaptation actions refer specifically to priority initiatives and measures for reducing adverse climate effects or increasing positive effects. Adaptation needs vary greatly according to place, population, and sector. Effective strategic adaptation planning focuses first on the systems most vulnerable²¹ to adverse climate impacts.

Actions were identified on the basis of observations recorded in recent decades in sectors that had conducted climate-change risk and vulnerability analyses.

Only the water, agriculture, and forestry sectors had analyzed the climate risks and vulnerabilities they face. On that basis, they identified adaptation measures incorporated into the NCP. The measures were identified through the "Climate Proofing" process²² shown in Figure 10.

Where relevant, lessons from experience at the regional and Mediterranean levels were used.

The proposed sectoral actions were grouped into two types of action projects:

- Sectoral action projects combining actions from the same sector; and
- Intersectoral action projects combining actions from several sectors.

The work conducted by institutions on adaptation showed that some actions would require implementation periods extending beyond 2025 because the implementation conditions for certain actions had not yet been defined.

The NCP therefore applies two time horizons to adaptation actions:

- The short-term action program for 2020-2025 addresses urgent needs and contains actions that can be carried out in the coming years. The human-resource capacity and skills needed for implementation already exist. The principal constraint is obtaining financial resources. The short-term program constitutes the NCP 2020-2025 action plan.

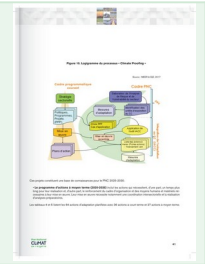
²¹ Climate vulnerability may be defined as the degree to which a system is capable or incapable of coping with adverse climate effects. Vulnerability depends on the character, magnitude, and rate of climate change and variation to which the system is exposed, its sensitivity, and its adaptive capacity (IPCC, 2007). It encompasses several concepts and elements, including susceptibility to harm and a lack of capacities to anticipate, cope, and recover. It has social, economic, environmental, institutional, cultural, and other dimensions. Vulnerability analyses are therefore an effective tool for identifying and prioritizing adaptation measures.

²² Climate Proofing is a methodological approach designed to integrate climate-related issues into development plans. The tool is based on climate-change principles established in the OECD guidance document on adaptation to climate change and development cooperation, as adopted by GIZ.

SOURCE PDF PAGE 41

FIGURE 10. FLOWCHART OF THE “CLIMATE PROOFING” PROCESS

English translation below; original source-page thumbnail at right.

**FIGURE 10. FLOWCHART OF THE “CLIMATE PROOFING” PROCESS**

Source: MEER & GIZ, 2017

Current Programming Framework

Sector Strategy

Policies, Programs, and Projects (PPP)

Implementation

Action Plans

NCP Framework

Preparation of the sectoral Risk and Vulnerability Analysis (ARV)

Identification of units exposed to climate change

Adaptation measures

Selection of PPP / application case

Application of the IACC tool - Integration of Climate-Change Adaptation

Targeted implementation

List of actions to be undertaken: action sheets and possible green financing

Adaptation measures

These projects form a knowledge base for the NCP 2025-2030.

The medium-term action program for 2020-2035 includes actions that require both a longer implementation period and strengthening of the organizational framework and the human and material resources needed for implementation. Their implementation requires, in particular, intersectoral coordination and preparatory analyses.

Tables 4 and 5 list the planned adaptation actions. The source text states that there are 64 actions, comprising 36 short-term actions and 27 medium-term actions; those two stated subtotals add to 63. This numerical inconsistency is retained and explicitly noted rather than silently resolved.

SOURCE PDF PAGE 42

TABLE 4. SHORT-TERM ADAPTATION ACTIONS, 2020-2025*English translation below; original source-page thumbnail at right.***TABLE 4. SHORT-TERM ADAPTATION ACTIONS, 2020-2025****Theme 1 - Coastal Risk Management****ACT 1** - Dredge ports. Lead: MTPT.**ACT 2** - Protect maritime infrastructure against erosion. Lead: MTPT.**Theme 2 - Risk Management and Marine Observation and Monitoring****ACT 3** - Wave forecasting. Lead: MTPT.**ACT 4** - Deploy radiotracers in the coastal area. Lead: MTPT.**ACT 5** - Develop a database on the Algerian coast and establish a catalog of sediment maps. Lead: MTPT.**ACT 6** - Establish a marine-monitoring network. Lead: MTPT.**ACT 7** - Observe the sea surface through satellite monitoring and remote sensing. Lead: ASAL.**ACT 8** - Collect data on acidification of marine waters. Lead: MADRP.**Theme 3 - Flood-Risk Management****ACT 9** - Prepare a Flood-Risk Prevention Plan (PPRI) for the Seybouse watershed. Lead: MRE.**ACT 10** - Establish a flood-warning system at the scale of the Seybouse watershed. Lead: MRE.**ACT 11** - Prepare and revise Master Plans for Development and Urban Planning (PDAU) and Land-Use Plans (POS). Lead: MHUV.**ACT 12** - Protect the infrastructure network against flooding. Lead: MTPT.**Theme 4 - Water Resources****ACT 13** - Assess climate-change impacts on coastal aquifers in Algeria and combat seawater intrusion. Lead: MRE.**ACT 14** - Assess future climate-change impacts on the regulatory capacity of reservoir dams. Lead: MRE.**ACT 15** - Optimize and rehabilitate irrigation systems. Lead: MRE.**ACT 16** - Develop and expand drip irrigation. Leads: MADRP and MRE.**Theme 5 - Agriculture****ACT 17** - Enhance and replicate the experience of Oran Wilaya in producing organic compost as a natural fertilizer. Lead: MEER.**ACT 18** - Establish a geographic information system for effective management of agricultural droughts. Lead: MADRP.**Theme 6 - Health: Observation and Monitoring****ACT 19** - Support implementation of childhood-diarrhea surveillance in Oran Wilaya. Lead: MSPRH.**ACT 20** - Support establishment of a pollen metrology monitoring system in the Annaba region. Lead: MSPRH.**ACT 21** - Assess toxicological risk from soil contamination associated with reuse of treated domestic wastewater for irrigation. Lead: MSPRH.**ACT 22** - Assess the chemical quality of water resources. Lead: MSPRH.**ACT 23** - Monitor and issue alerts concerning climate-related health risks. Lead: MSPRH.**ACT 24** - Support establishment of a unit dedicated to microbiological monitoring of aquatic environments. Lead: MSPRH.**ACT 25** - Support completion of a level-2 containment insectarium (I2) at the Pasteur Institute of Algeria. Lead: MSPRH.**ACT 26** - Protect emergency departments from heat waves by installing suitable air-conditioning systems. Lead: MSPRH.**ACT 27** - Provide information and communication on climate-change impacts on human health. Lead: MSPRH.

SOURCE PDF PAGE 43

TABLE 4. SHORT-TERM ADAPTATION ACTIONS, 2020-2025 - CONTINUED*English translation below; original source-page thumbnail at right.***TABLE 4. SHORT-TERM ADAPTATION ACTIONS, 2020-2025 - CONTINUED****Theme 7 - Erosion, Desertification, the Green Dam, and Soil Protection****ACT 28** - Restore rangelands through temporary closure to grazing. Lead: MADRP.**ACT 29** - Combat sand encroachment and equip exposed areas with specific response resources. Lead: MADRP.**Theme 8 - Vulnerability of Local Authorities****ACT 30** - Prepare local adaptation plans for three pilot wilayas: M'Sila, El Bayadh, and Sidi Bel Abbès. Lead: MEER.**ACT 31** - Strengthen local disaster-risk-management capacity in all 48 wilayas. Lead: MICLAT.**ACT 32** - Establish a network of resilient cities. Lead: MICLAT.**ACT 33** - Assist in formulating urban-resilience master plans for Chlef and Oran. Lead: MICLAT.**ACT 34** - Map natural disasters and their impacts on strategic road, port, and airport infrastructure. Lead: MTPT.**ACT 35** - Analyze climate variability and project heat waves in major coastal cities. Lead: IHFR.**ACT 36** - Study the evolution of drought in western Algeria for the 2050 and 2100 horizons. Lead: IHFR.**TABLE 5. MEDIUM-TERM ADAPTATION ACTIONS, 2020-2035****Theme 1 - Infrastructure****AMT 1** - Equip areas exposed to snowfall with effective snow-removal resources. Lead: MTPT.**AMT 2** - Equip areas exposed to sand encroachment with sand-removal resources. Lead: MTPT.**AMT 3** - Adapt infrastructure in southern Algeria to extreme heat. Lead: MTPT.**AMT 4** - Strengthen transport research related to climate change. Lead: MTPT.**Theme 2 - Marine Biodiversity****AMT 5** - Observe and analyze marine-ecosystem dynamics (DIMA). Lead: MADRP.**AMT 6** - Submerge artificial reefs to restore fishing areas. Lead: MADRP.**AMT 7** - Establish a network to monitor climate-change impacts on Posidonia seagrass beds. Lead: MEER.**AMT 8** - Establish a national program for managing invasive aquaculture species. Lead: MADRP.**AMT 9** - Establish a coralligenous-habitat monitoring network. Lead: MEER.**Theme 3 - Water Resources****AMT 10** - Beneficially use nonconventional water resources in arid areas. Lead: MRE.**AMT 11** - Conduct a national study on reuse of treated wastewater. Lead: MRE.**AMT 12** - Reuse demineralized drainage water in agriculture in the Oued Righ Valley. Lead: MRE.**AMT 13** - Implement a pilot aquifer-recharge project using surface-water resources. Lead: MRE.**AMT 14** - Study the vulnerability of groundwater resources to climate-change effects. Lead: MRE.

SOURCE PDF PAGE 44

TABLE 5. MEDIUM-TERM ADAPTATION ACTIONS, 2020-2035 - CONTINUED*English translation below; original source-page thumbnail at right.***TABLE 5. MEDIUM-TERM ADAPTATION ACTIONS, 2020-2035 - CONTINUED****Theme 4 - Agriculture****AMT 15** - Improve agricultural practices for better use of rainwater in semi-arid areas. Lead: MADRP.**AMT 16** - Introduce and beneficially use new adapted species, such as Opuntia. Lead: MADRP.**AMT 17** - Select and genetically improve varieties of cereals, food legumes, forage crops, vegetable crops, and fruit trees. Lead: MADRP.**AMT 18** - Develop direct-seeding techniques in marginal and erosion-sensitive areas. Lead: MADRP.**AMT 19** - Preserve the Bouhmama apple orchards against advanced drought and establish measures for sustainable irrigation. Lead: MADRP.**Theme 5 - Terrestrial Biodiversity****AMT 20** - Restore the steppe ecosystem through regeneration of alfa grass. Lead as printed: MADPR. [Elsewhere in the document, the acronym appears as MADRP.]**AMT 21** - Establish an observatory for ecological, climatic, and socioeconomic monitoring of the steppe ecosystem. Lead as printed: MADPR.**AMT 22** - Adapt forest ecosystems to climate change and strengthen their ecological resilience, focusing on high-altitude summit species. Lead as printed: MADPR.**AMT 23** - Strengthen wetland resilience to climate change. Lead as printed: MADPR.**AMT 24** - Strengthen the climate resilience of protected areas. Lead as printed: MADPR.**AMT 25** - Prepare an integrated development plan for the Bibans massif, covering the Ighil Ali area in Béjaïa and Theniet Enassr in Bordj Bou Arréridj. Lead: MEER.**AMT 26** - Establish protected areas for fisheries. Lead as printed: MADPR.**AMT 27** - Enhance and improve the climate resilience of oasis populations and landscapes in Algeria. Lead: MEER.**5.3. Mitigation**

An activity is classified as a climate-change mitigation action if it contributes to the mitigation objective of stabilizing GHG concentrations in the atmosphere at a level that prevents dangerous anthropogenic interference with the climate system, as stated in Article 2 of the UNFCCC. An activity is considered mitigation if it directly contributes to reducing emissions through one of the following:

Reducing or stabilizing GHG emissions in energy, transport, industry, and agriculture by using new forms of energy or renewable energy; measures to improve the efficiency of existing generators, machinery, and equipment; or energy-demand-management tools.

Reducing methane emissions through waste management and wastewater treatment.

Developing, transferring, and promoting technologies and know-how, and building capacities to control, reduce, or prevent anthropogenic GHG emissions, particularly in waste management, transport, energy, agriculture, and industry.

Protecting and improving GHG sinks and reservoirs through sustainable forest management, afforestation and reforestation, and restoration of areas affected by drought and desertification.

Tables 6-13 list the 76 mitigation actions in industry (17 actions), waste (7), energy (19), forestry (3), transport (15), local authorities (4), buildings and housing (10), and tourism (1).

SOURCE PDF PAGE 45

TABLE 6. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - INDUSTRY*English translation below; original source-page thumbnail at right.***TABLE 6. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - INDUSTRY**

- 1 - Renovate and restart two synchronous compensators at the Sider El Hadjar complex. Lead: MIM.
- 2 - Reduce coke consumption in the blast furnace by optimizing process operating parameters at Sider El Hadjar. Lead: MIM.
- 3 - Strengthen the complex's capacity to improve energy performance and reduce GHG and air-pollutant emissions at Sider El Hadjar. Lead: MIM.
- 4 - Use solar photovoltaic power for lighting at the Sider El Hadjar complex. Lead: MIM.
- 5 - Modernize the ignition system for ladle-heating stations at the two steelworks and reinject hot steelworks gases to heat furnace ladles at Sider El Hadjar. Lead: MIM.
- 6 - Install variable-speed drives on high-power motors to reduce energy consumption in production units at Sider El Hadjar. Lead: MIM.
- 7 - Improve overall energy management in the production units of the Sider El Hadjar complex; optimize and renovate facilities to reduce energy consumption. Lead: MIM.
- 8 - Install power-factor-correction capacitor banks and return the turbo-alternator to service to reduce consumption and generate electricity on site. Lead: MIM.
- 9 - Install new high-energy-efficiency induction furnaces in ferrous- and nonferrous-metal foundries at El Harrach and Tiaret. Lead: MIM.
- 10 - Install solar LED systems for interior and exterior lighting at ENAP units. Lead: MIM.
- 11 - Convert ALFAPIPE service vehicles to LPG. Lead: MIM.
- 12 - Co-incinerate high-calorific-value household and industrial waste in cement kilns at GICA units. Lead: MIM.
- 13 - Modernize facilities and replace cupola furnaces with electric induction furnaces for metal melting at ALFEL. Lead: MIM.
- 14 - Optimize use of the "zinc pellet" transformer to reduce energy consumption at ALZINC's zinc-production unit. Lead: MIM.
- 15 - Renew and modernize the ALFET foundry in Tiaret. Lead: MIM.
- 16 - Install an ACS 800 variable-speed motor drive in the process to reduce energy consumption at ALZINC. Lead: MIM.
- 17 - Provide support and technical assistance to industrial enterprises for modernization and upgrading of facilities to conserve resources and improve environmental performance. Lead: MIM.

TABLE 7. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - WASTE

- 18 - Develop value-recovery pathways for sludge from wastewater-treatment plants through co-incineration in cement kilns. Lead: MIM.
- 19 - Dismantle and recover value from waste electrical and electronic equipment. Lead: MEER.
- 20 - Promote green-economy entrepreneurship in the Algerian wilayas listed in the source: Bordj Bou Arréridj, Tizi Ouzou, Illizi, Batna, Blida, and Aïn Defla. Lead: MEER. [The source says "five wilayas" but lists six; this inconsistency is retained.]
- 21 - Eliminate illegal dumps and recover energy from closed cells at engineered landfill centers. Lead: MEER.
- 22 - Conduct a market study on industrial-waste recovery within Algeria's circular economy. Lead: MEER.
- 23 - Recycle and recover waste and produce energy. Lead: MEER.
- 24 - Implement a national pilot project for low-GHG-emission recovery of household and similar waste. Lead: MIM.

SOURCE PDF PAGE 46

TABLE 8. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - ENERGY*English translation below; original source-page thumbnail at right.***TABLE 8. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - ENERGY**

- 25** - Develop Algeria's solar-water-heater market by installing 2,000 individual solar water heaters in households nationwide. Lead: MDE.
- 26** - Support widespread adoption of efficient LED lighting in Algerian homes. Lead: MDE.
- 27** - Strengthen the energy-transition market by converting 500,000 light gasoline vehicles to LPG. Lead: MDE.
- 28** - Develop high-energy-efficiency combined-cycle power plants in the SPE/Sonelgaz fleet. Lead: MDE.
- 29** - Hybridize diesel power plants in southern Algeria by installing additional solar photovoltaic plants with a total capacity of 50 MW. Lead: MDE.
- 30** - Implement a systematic preventive-control program to reduce SF₆ emissions from Sonelgaz electrical equipment and installations. Lead: MDE.
- 31** - Strengthen the capacity of Sonelgaz companies to monitor the network and reduce emissions from incidents and natural-gas leaks nationwide. Lead: MDE.
- 32** - Update mapping and monitoring of the national natural-gas distribution network. Lead: MDE.
- 33** - Implement national programs to extend the public natural-gas distribution network nationwide. Lead: MDE.
- 34** - Supply solar energy to 2,897 isolated homes through individual photovoltaic kits. Lead: MDE.
- 35** - Control natural-gas leaks from the gas-transmission network using laser technology. Lead: MDE.
- 36** - Renovate oil-production facilities to reduce emissions and flaring in oil fields. Lead: MDE.
- 37** - Substitute energy sources for equipment and sites. Lead: MDE.
- 38** - Establish energy-management systems, obtain ISO 50001 certification, and audit energy-management systems at production facilities. Lead: MDE.
- 39** - Maintain facilities to reduce emissions and protect equipment. Lead: MDE.
- 40** - Modernize, renew, and rehabilitate gas-industry facilities to control and reduce gaseous atmospheric emissions from the GL1Z and GL2Z natural-gas-liquefaction complexes at Arzew. Lead: MDE.
- 41** - Strengthen the capacity of Sonatrach structures and units to conduct internal energy audits of energy-production sites and facilities. Lead: MDE.
- 42** - Install new systems to recover associated gas from oil fields and natural-gas production. Lead: MDE.
- 43** - Deploy solar photovoltaic energy for water pumping and irrigation systems on farms in southern Algeria. Lead: MEER.

TABLE 9. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - FORESTRY

- 44** - Explore forest carbon-sequestration potential, using the Senalba state forest in Djelfa Wilaya as a case study. Lead: MADRP.
- 45** - Implement an integrated arid-land-restoration project in the Green Dam zone in the context of climate change. Lead: MADRP.
- 46** - Develop a forest-fire strategy to reduce fires and develop prevention and management tools. Lead: MADRP.

SOURCE PDF PAGE 47

TABLE 10. PROPOSED ACTIONS AND ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - TRANSPORT*English translation below; original source-page thumbnail at right.***TABLE 10. PROPOSED ACTIONS AND ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - TRANSPORT**

- 47 - Implement an onboard Electronic Flight Bag Class 2 information-management system. Lead: MTPT.
- 48 - Implement a Fuel Dashboard onboard Air Algérie aircraft. Lead: MTPT.
- 49 - Install the Electronic Flight Folder onboard Air Algérie aircraft. Lead: MTPT.
- 50 - Acquire ground power units and air-conditioning units to reduce kerosene consumption on the ground. Lead: MTPT.
- 51 - Optimize air navigation. Lead: MTPT.
- 52 - Convert vehicles to LPG and acquire new LPG vehicles at airports. Lead: MTPT.
- 53 - Install photovoltaic panels for lighting at Biskra Airport. Lead: MTPT.
- 54 - Install LED perimeter-lighting systems at Algerian airports. Lead: MTPT.
- 55 - Install solar lighting at Hassi Messaoud Airport. Lead: MTPT.
- 56 - Develop and strengthen the rail network. Lead: MTPT.
- 57 - Convert airport service-vehicle fleets to LPG. Lead: MTPT.
- 58 - Modernize port equipment, integrate solar energy, and improve energy efficiency at the ports of Arzew, Ghazaouet, Béjaïa, and Mostaganem. Lead: MTPT.
- 59 - Modernize urban public transport by extending Algiers Metro Line 1 and constructing Oran's first metro line. Lead: MTPT.
- 60 - Modernize urban public transport by constructing tramways in Algerian cities. Lead: MTPT.
- 61 - Construct cable-transport systems in Algiers and Tizi Ouzou and modernize the Oran gondola lift. Lead: MTPT.

TABLE 11. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - LOCAL AUTHORITIES

- 62 - Modernize public-lighting management, generalize LED lamps, and develop solar lighting. Lead: MICLAT.
- 63 - Generalize LED lighting and install solar equipment, while optimizing and standardizing lighting in schools and cafeterias. Lead: MICLAT.
- 64 - Generalize LED lighting and install solar equipment, while optimizing and standardizing lighting and air conditioning according to use in mosques and Quranic schools. Lead: MICLAT.
- 65 - Generalize LED lighting and install solar equipment, while optimizing and standardizing lighting in municipal buildings and public infrastructure. Lead: MICLAT.

SOURCE PDF PAGE 48

TABLE 12. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - BUILDINGS AND HOUSING

English translation below; original source-page thumbnail at right.

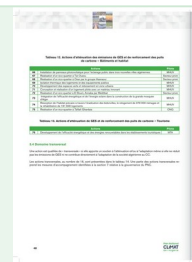


TABLE 12. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - BUILDINGS AND HOUSING

- 66** - Install photovoltaic panels for public lighting in three new Algerian cities. Lead: MHUV.
- 67** - Develop an eco-district in Tizi Ouzou. Lead: Private sector.
- 68** - Develop an eco-district in Oran through the Hasnaoui Group. Lead: Private sector.
- 69** - Thermally insulate housing and public facilities. Lead: MHUV.
- 70** - Develop green spaces and reforest urban areas. Lead: MHUV.
- 71** - Design and construct a pilot dwelling using an innovative material. Lead: MHUV.
- 72** - Develop an eco-district in El Bouni, Annaba, through Méditbat. Lead: Private sector.
- 73** - Integrate energy efficiency and solar energy into construction of the Great Mosque of Algiers. Lead: MHUV.
- 74** - Eliminate precarious housing through eradication of slums, rehousing of 379,000 households, and rehabilitation of 181,848 homes. Lead: MHUV.
- 75** - Develop an eco-district in Tafilalt, Ghardaïa. Lead: NGO.

TABLE 13. ACTIONS TO MITIGATE GHG EMISSIONS AND STRENGTHEN CARBON SINKS - TOURISM

- 76** - Develop energy efficiency and renewable energy in tourism establishments. Lead: MTA.

5.4. Cross-Cutting Area

An action is classified as cross-cutting when it supports mitigation and/or adaptation even though it does not itself reduce GHG emissions or directly contribute to adapting Algerian society to climate change.

The 16 cross-cutting actions are presented in Table 14. Some repeat the supporting measures identified in Section 7 on NCP governance.

SOURCE PDF PAGE 49

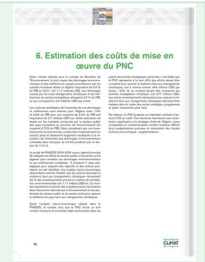
TABLE 14. SHORT-TERM CROSS-CUTTING ACTIONS, 2020-2025*English translation below; original source-page thumbnail at right.***TABLE 14. SHORT-TERM CROSS-CUTTING ACTIONS, 2020-2025**

- 1 - Establish the national GHG-emissions inventory system. Lead: MEER.
- 2 - Strengthen national institutions on climate-change issues. Lead: MEER.
- 3 - Establish an MRV system for NCP actions. Lead: MEER.
- 4 - Prepare a national climate-change research program and integrate climate-change research into other existing research programs. Lead: MEER.
- 5 - Integrate climate change into school and university curricula. Lead: MEER.
- 6 - Strengthen technological and scientific monitoring of climate-adaptation and mitigation measures. Lead: MEER.
- 7 - Strengthen financial institutions' capacity in climate finance. Lead: MEER.
- 8 - Develop new mechanisms to finance climate-adaptation and mitigation measures. Lead: MEER.
- 9 - Conduct climate-change awareness and information campaigns. Lead: MEER.
- 10 - Install an atmospheric station to measure short-lived climate pollutants. Lead: MEER.
- 11 - Establish an accredited testing laboratory for quality control of solar water heaters. Lead: MEER.
- 12 - Establish an accredited laboratory for quality control of photovoltaic modules. Lead: MEER.
- 13 - Conduct renewable-energy research projects with socioeconomic impacts. Lead: MEER.
- 14 - Establish an accredited metrology laboratory for calibrating pyranometers. Lead: MEER.
- 15 - Integrate climate change into the industrial-zone rehabilitation program. Lead: MEER.
- 16 - Develop new industrial zones that are resilient to climate-change effects. Lead: MEER.

SOURCE PDF PAGE 50

6. ESTIMATED COST OF IMPLEMENTING THE NCP

English translation below; original source-page thumbnail at right.

**6. ESTIMATED COST OF IMPLEMENTING THE NCP**

According to a study carried out for the Ministry of Environment, the average annual cost of environmental damage and inefficiencies caused by human activities in Algeria reached the equivalent of 6.9% of GDP in 2017, or USD 11.7 billion. Damage caused solely by climate change and inefficiencies in the energy sector amounted to 2.9% of GDP, equivalent to USD 4.9 billion per year.

The costs of remedying all such damage and inefficiencies in Algeria are estimated at between 1.4% and 2.9% of GDP, averaging 2.2% of GDP, or the equivalent of USD 3.7 billion per year. This estimate is based on the amounts European public sectors devote to the environment - an average of 0.75% of GDP. Because those investments do not include private-sector investment and remain far below what would be required to remedy all observed environmental damage in those countries, they were weighted by a factor of 1.5 to 2.

The draft PNAEDD 2020-2024 has the principal objective of catalyzing Algerian public- and private-sector efforts to remedy observed environmental damage and inefficiencies. It contains seven strategic pillars for which priority objectives and actions were identified. A top-down macroeconomic analysis indicates that actions promoting resilience to climate change require 30% of annual environmental-remediation investment, or USD 1.11 billion per year. This amount represents the combined investment required in the national economy from the Government and its public- and private-sector partners to ensure the country's climate resilience.

According to the macroeconomic analysis used in the PNAEDD, and given that the NCP includes many actions and activities already mentioned in other sectoral strategic documents, the NCP alone represents 40% of the effort required to ensure climate resilience. This amounts to at least approximately USD 444 million per year. Of that amount, 62% - USD 277 million - should be devoted to climate-adaptation activities. Other investments needed for climate resilience should be made under the other strategies, programs, and plans mentioned above.

The NCP also proposes a coherent set of 155 actions. They make a significant contribution to Algeria's climate strategy. Their complexity and multisectoral nature, however, make precise budgeting difficult and require additional technical and economic studies.

SOURCE PDF PAGE 51

7. GOVERNANCE OF THE NCP

English translation below; original source-page thumbnail at right.



7. GOVERNANCE OF THE NCP

Section 7 sets out arrangements for preparing, implementing, monitoring, and evaluating the National Climate Plan. It proposes a governance framework that defines the responsibilities of the different stakeholders and identifies arrangements for monitoring and evaluating implementation and for updating the NCP.

Section 7 identifies the NCP-related tasks and institutions concerned; addresses climate-change stakeholders; describes the NCP governance framework; and provides institutional capacity-building measures intended to create conditions favorable to achieving Algeria's climate-strategy objectives.

7.1. Tasks Arising from the NCP

Formulating the NCP, implementing its actions, and monitoring, evaluating, and updating them require the following:

- Ensure interinstitutional collaboration and coordination among sectors concerned with adaptation and mitigation in plan design, implementation, and action monitoring and evaluation.

- Strengthen integration of climate change into all national and sectoral policies, taking account of territorial programming such as the SNAT, wilaya development plans, and local plans.

- Plan financial needs for implementing actions, establish appropriate financing mechanisms, and incorporate international financing instruments into regulations for developing climate-mitigation or adaptation projects.

- Establish an action monitoring and evaluation system.

- For mitigation actions, develop a functional national Measurement, Reporting, and Verification (MRV) system to assess and verify implementation and results, measure compliance with GHG-reduction commitments, assess associated financial costs, and report results.

- Develop a monitoring and evaluation system for adaptation actions to measure implementation levels and resulting impacts.

- Provide information and communication on national climate policy to all stakeholders, enabling regular and objective evaluation of the NCP.

- Plan regular updates of the NCP to ensure consistency with changes in national development policy; scientific and technical climate knowledge; technological development; mitigation opportunities; and the financing available at the time.

7.2. Climate-Change Stakeholders

Algeria has several bodies capable of assuming the roles described above. They are principally institutions whose statutory mandates concern climate change: the CNC; the MEER Climate Change Directorate (DCC) and its two sub-directorates responsible for mitigation and adaptation; ANCC; and ONEDD.

Several ministries have also begun integrating climate change into their strategies and have structures for which reducing GHG emissions and adapting to climate change are important concerns. These include the Ministry of Energy through APRUE; the Ministry of Agriculture, Rural Development, and Fisheries; and the Ministry of Water Resources. The list continues on the next source page.

SOURCE PDF PAGE 52

7.2. Climate-Change Stakeholders - Continued

English translation below; original source-page thumbnail at right.

**7.2. Climate-Change Stakeholders - Continued**

Other relevant bodies include the National Meteorological Office (ONM); National Center for Cleaner Production Technologies (CNTPP); Renewable Energy Development Center (CDER); Algerian Space Agency (ASAL); General Directorate of Forests (DGF); national institutes for agronomic research (INRAA), forestry research (INRF), and meteorology and climatology research (IHFR); and other national organizations.

7.3. NCP Governance Framework

Linking existing roles and stakeholders makes it possible to formulate a governance framework in which existing institutions can cover all identified functions.

7.3.1. Preparation

The CNC prepares the NCP. Its role is to formulate the Plan and ensure consistency with Algeria's strategic documents and international commitments. The CNC selects and prioritizes actions for implementation and determines the document's structure and content.

MEER chairs and organizes the CNC's work and, through the DCC, provides the CNC secretariat. It prepares statutory CNC meetings, facilitates the Committee's work, and prepares minutes of meetings and sessions. ANCC supports the CNC's work with scientific and technical expertise.

The CNC's intersectoral composition ensures representation of all stakeholders concerned with climate change. It is the forum for exchange and consultation that secures sectoral ownership of the actions for which sectors are responsible and collaboration among sectors on joint actions.

Outside CNC meetings, the MEER Climate Change Directorate provides the link, exchanges, and intersectoral collaboration.

The CNC acts as a bridge to sectors, civil society, and associations.

7.3.2. Implementation of Actions

The NCP does not establish new arrangements for managing public policies. Implementation of actions is entirely the responsibility of the institutional structures that participated in preparing the NCP.

Structures responsible for implementation are the sectoral institutions responsible for carrying out sector policies: ministries, agencies and specialized institutions, supervised bodies, wilaya technical directorates, and decentralized and local technical services. They are responsible for:

- Implementing NCP actions according to their institutional and regulatory mandates and competencies;
- Establishing the financial arrangements needed to fund actions;
- Defining implementation arrangements and resources and implementation schedules; and
- Monitoring and evaluating action results and preparing reports for submission to higher-level climate-change bodies.

Intersectoral leadership for each action rests with the sector principally concerned, designated the action lead. The lead coordinates activities carried out by partners involved in the action project and reports first to its sector focal point represented on the CNC.

With support from its mitigation and adaptation sub-directorates, the DCC monitors NCP implementation by coordinating and facilitating intersectoral collaboration. The sub-directorates assist sectors in identifying potential national and international financing sources and following the related procedures.

The ANCC Scientific and Technical Council contributes before and after implementation through analysis and recommendations concerning projects submitted to it.

SOURCE PDF PAGE 53

7.3.3. Communication

English translation below; original source-page thumbnail at right.



7.3.3. Communication

Successful NCP implementation depends on continuous and active communication to promote acceptance and ownership of NCP actions among all stakeholders, including the population and development partners.

The DCC communicates climate policy in consultation with the sectors leading NCP actions. In consultation with the CNC, the DCC must prepare a communication plan specifying objectives, target audiences, and the means to be used.

Communication should address every segment of society: national decision-makers, local officials, administrations, organizations, businesses, civil society and associations, children, students, and the public in general. Particular attention should be given to stakeholders directly affected by or involved in implemented actions.

7.3.4. Monitoring and Evaluation

Monitoring and evaluation play an important role in implementation. They make it possible to:

- Report NCP progress at regular intervals, make implementation visible, and identify lessons learned and good practices;
- Identify difficulties and barriers and remedy identified shortcomings, gaps, and needs; and
- Formulate recommendations to close gaps, remove barriers, and overcome difficulties so as to ensure continuity and continued NCP implementation.

For monitoring purposes, each sector responsible for implementing NCP actions must collect information on the progress of actions and projects and prepare periodic reports. This information must populate the action-result and performance indicators identified in the action sheets. It is transmitted to the DCC for consolidation.

The DCC leads monitoring, centralizes information, evaluates indicators, and produces regular reports describing implementation levels and performance. Reports are transmitted to the CNC, which must be informed regularly of action progress and may take necessary adjustment or redirection measures.

The CNC determines monitoring arrangements - resources and deadlines - and tools, including a dashboard listing specific objectives, expected results, and result and performance indicators. It may decide on data sources and collection, the methodology followed, constraints, and monitoring processes. As part of its mandate, the CNC prepares a report for submission to the Government.

With respect to the international community, an MRV system provides transparent measurement and verification of GHG reductions. It also assures funders that planned NCP actions are being implemented properly. Countries seeking international finance must be able to plan financing needs, measure implementation and compliance with commitments, assess associated financial costs, report, and verify application of standards and good practices through MRV.

Countries with an MRV system recognized as transparent by the international community therefore have greater credibility with funders and are better placed to attract financing for their NDCs.

Monitoring NCP implementation requires two tools:

- A national GHG-emissions inventory system that measures mitigation efforts and estimates GHG emissions and removals from all activities within the national territory; and
- The NCP MRV system, described on the next source page.

SOURCE PDF PAGE 54

7.3.4. Monitoring and Evaluation - Continued

English translation below; original source-page thumbnail at right.

**7.3.4. Monitoring and Evaluation - Continued**

The NCP MRV system reports on the degree of action implementation, results, and performance. The DCC leads establishment of the national GHG-inventory system, drawing on all necessary national and international expertise. ANCC compiles the national GHG inventory in collaboration with sectoral entities concerned with inventories.

Beyond monitoring, objective evaluation of the NCP must be planned and entrusted to an external body not directly involved in preparing or implementing the Plan.

7.3.5. Financing

Sectors sponsoring actions are responsible for financing implementation. They must budget the actions or arrange financing with national or international funders.

Under MEER's responsibility, an ad hoc committee composed of representatives of MEER, the Ministry of Finance, and the Ministry of Foreign Affairs should be established to coordinate financing opportunities with international funders.

Although allocating more national resources to adaptation and mitigation actions is conceivable, recent economic conditions make mobilization of new financial resources appear necessary. Two principal sources can be mobilized:

- International funds and bilateral and multilateral cooperation programs; and
- National funds and innovative national financing mechanisms.

7.3.6. Updating the NCP

Because knowledge is developing and climate change and its impacts are evolving, updates of the NCP must be planned.

Updates will take account of progress achieved; new adaptation and mitigation issues and challenges; the socioeconomic context; the policy environment; and the State's national and international commitments.

The CNC determines the updating arrangements through:

- Establishing a principle of periodic revision every five years, coinciding with the end of the NCP's first five-year period; and
- Taking account in the revision of results achieved by the first NCP; effectiveness of implemented measures; observed differences from initial climate projections; changes in the vulnerability of development sectors; mitigation efforts achieved; and changes in Algeria's international commitments, including the NDC.

Table 15 provides an overview of NCP governance and the allocation of responsibilities to each entity described above. Annex 1 proposes an approach for operationalizing the NCP.

SOURCE PDF PAGE 55

TABLE 15. OVERVIEW OF THE NCP GOVERNANCE FRAMEWORK

English translation below; original source-page thumbnail at right.

**TABLE 15. OVERVIEW OF THE NCP GOVERNANCE FRAMEWORK**

Task - Responsibility - Support

Preparation of the NCP and consultation with the Government

Responsibility: CNC

Support: MEER, as CNC secretariat

Implementation of actions

Responsibility: Ministries and institutions responsible for the actions

Support: MEER

Communication concerning the NCP

Responsibility: MEER prepares and implements the strategy

Support: Sectors

Monitoring and evaluation

Responsibility: Sectors; CNC defines deadlines and the monitoring and evaluation process; MEER conducts monitoring and prepares monitoring reports

Support: MEER

Financing

Responsibility: Sectors responsible for the actions

Support: MEER

Updating

Responsibility: CNC

Support: Ministries and institutions responsible for the actions

7.4. Supporting Measures

Establishing and operating the governance framework requires a series of supporting measures to create favorable conditions for implementation, monitoring, evaluation, and updating.

The CNC must be strengthened so it can perform its functions fully, in particular through:

Expanding its composition to include other economic stakeholders and civil society;

Training members in strategic climate-policy planning; and

Strengthening the CNC secretariat with two staff members assigned to MEER's Climate Change Directorate.

For implementation, it is necessary to:

Strengthen sectoral skills to facilitate identification of relevant actions and definition of implementation arrangements; and

Strengthen intersectoral coordination through the DCC.

For monitoring and evaluation:

Develop appropriate methodologies and tools and train responsible MEER personnel and all stakeholders involved in NCP implementation and monitoring.

For financing:

Strengthen the capacity of DCC personnel, other MEER managers, and focal points for international funds such as the GEF and GCF so they can fulfill their roles with funders; and

Create an ad hoc climate-finance group composed of MEER, Ministry of Finance, and Ministry of Foreign Affairs representatives to coordinate international financing opportunities.

Together with the communications directorate and MEER's supervised bodies, the DCC should prepare an annual NCP communication program to disseminate and explain the Plan to the various stakeholders and actors, including the general population.

SOURCE PDF PAGE 56

8. REFERENCES

English translation below; original source-page thumbnail at right.

**8. REFERENCES**

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SOURCE PDF PAGE 57

8. REFERENCES - CONTINUED

English translation below; original source-page thumbnail at right.



8. REFERENCES - CONTINUED

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SOURCE PDF PAGE 58

9. ANNEXES

English translation below; original source-page thumbnail at right.

**9. ANNEXES****ANNEX 1: MONITORING FRAMEWORK AND MONITORING TOOLS****1. Institutional Framework for Monitoring and Evaluating the NCP**

The framework should provide for the following bodies and processes:

- Annual NCP review;
- Sectoral reviews; and
- Territorial reviews.

The annual CNC-DCC review validates draft annual performance reports, the NCP five-year evaluation report for 2020-2025, and the draft performance matrix for future years.

Sectoral reviews are held twice a year, and their reports feed into the annual NCP review. They comprise a midyear sectoral review and an annual sectoral review. The midyear review examines the year's interim results and actions to correct shortcomings observed in NCP implementation.

Territorial reviews formally validate actions in local territorial climate plans and assess their implementation. Like sectoral reviews, at least two sessions are held each year, and validated reports feed into the CNC annual review. For NCP monitoring, regional reviews will verify that climate change is properly taken into account in local development plans, including regional and municipal development plans.

2. Monitoring Tools**2.1. National GHG-Emissions Inventory System - Mitigation Component**

To ensure its MRV status, the national GHG-emissions inventory system must include:

- Data and information on GHG emissions, mitigation and adaptation actions, and support requested and received, including financial flows, technology transfers, and skills development;
- Adoption of a standard format for compiling information; and
- Periodic analysis and evaluation of measures to ensure their reliability and compliance with established principles.

Preparation of an inventory must follow IPCC good practices (2006), particularly with respect to completeness, accuracy, consistency, and comparability.

A GHG-emissions inventory system is being established in Algeria in connection with preparation of the Second National Communication. The system must rest on a permanent institutional and technical structure.

2.2. Monitoring System

The system aims to monitor and facilitate evaluation of NCP actions in relation to the objectives pursued. It must collect, process, and disseminate information needed to monitor the progress and effects of adaptation actions through result and performance indicators, as provided in the template.

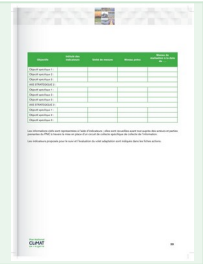
Implementation indicators measure the degree to which an action has been carried out. Performance indicators monitor results achieved. Together, the two types of indicators should determine the program's ability to achieve specific NCP climate-adaptation objectives.²³

²³ Each indicator is associated with a starting, or baseline, value and an objective, or target, value. During program implementation, the indicator's current value at time t - the achieved value - can be compared with its baseline and target values.

SOURCE PDF PAGE 59

ANNEX 1: MONITORING FRAMEWORK AND MONITORING TOOLS - CONTINUED

English translation below; original source-page thumbnail at right.



ANNEX 1: MONITORING FRAMEWORK AND MONITORING TOOLS - CONTINUED

Monitoring Template

Columns:

Objectives
Indicator Title
Unit of Measurement
Planned Level
Level Achieved as of [date]

Rows shown in the source template:

Specific Objective 1
Specific Objective 2
Specific Objective 3

• STRATEGIC PILLAR 2

Specific Objective 1
Specific Objective 2
Specific Objective 3

• STRATEGIC PILLAR 3

Specific Objective 1
Specific Objective 2
Specific Objective 3

Key information is represented through indicators. It is collected primarily from NCP stakeholders through establishment of a dedicated information-collection pathway.

The proposed indicators for monitoring and evaluating the adaptation component are provided in the action sheets.

SOURCE PDF PAGE 60

ANNEX 2: LIST OF CONTRIBUTORS

English translation below; original source-page thumbnail at right.



ANNEX 2: LIST OF CONTRIBUTORS

1. Technical Coordination

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Mr. Nouar LAIB - Director General for Environment and Sustainable Development

Ms. Fazia DAHLAB - Director of Climate Change

German Cooperation - GIZ Algeria

Ms. Charlotte SIEGERSTETTER - Head of Mission

2. Experts

Prof. Ménouër BOUGHEDAUI - Professor, University of Blida 1

Prof. Mahi TABET-AOUL - National Expert

Prof. David MARADAN - International Expert

3. National Climate Committee

Mr. Nouar LAIB - Ministry of Environment and Renewable Energy

Mr. Djihad Eddine BELKAS - Ministry of Foreign Affairs

Ms. Ghalia BENZIOUCHE - Ministry of Foreign Affairs

Mr. Samir BOUZID - Ministry of National Defense

Mr. Lazhar SOUALEM - Ministry of Foreign Affairs

Mr. Kamel KAILI - Ministry of the Interior, Local Authorities, and Spatial Planning

Mr. Amar LEULMI - Ministry of Finance

Ms. Zohra BOUHOUCHE - Ministry of Energy

Mr. Mohamed ZEGHDANI - Ministry of Religious Affairs and Wakfs

Ms. Samia MEZAIB - Ministry of National Education

Mr. Nour Eddine YASSA - Ministry of Higher Education and Scientific Research

Mr. Yacine ENNEHAITI - Ministry of Industry and Mines

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Mr. Hacene SID AHMED - Ministry of Commerce

Ms. Sabrina ATMANI - Ministry of Communication

Mr. Abdelkrim REZZAL - Ministry of Public Works and Transport

Mr. Tahar AICHAOUI - Ministry of Water Resources

Mr. Abdelhamid TERGUINI - Ministry of Tourism and Handicrafts

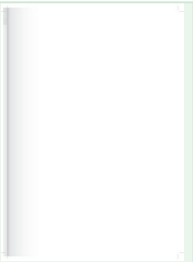
Ms. Saida BADREDDINE - Ministry of Health, Population, and Hospital Reform

Ms. Hamida MERABET - National Economic and Social Council

SOURCE PDF PAGE 61

Translated source page

English translation below; original source-page thumbnail at right.

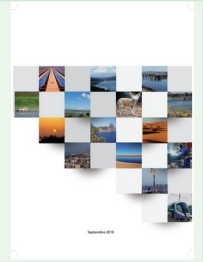


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BACK COVER

English translation below; original source-page thumbnail at right.



BACK COVER

September 2019

[The source back cover contains a photographic mosaic depicting solar infrastructure, reservoirs and wetlands, wildlife, desert and coastal landscapes, an urban area, industry, and public transport.]